

Deriving Born's Rule from Infinitesimal Structure

A Projective Selection Theorem for the Quantum Measurement Problem

Kenneth B. Reidbord

Independent Researcher, United States

Abstract

This paper derives the Born rule. The squared-amplitude weights that quantum theory assigns to measurement outcomes with extraordinary precision, but has never explained, are obtained here from infinitesimal fiber structure in the Plenum Field, an ordered field in which the real line arises as a quotient and each real point retains an underlying fiber of distinctions erased by the standard-part map. The single-click selection law

$$P(i \mid \text{click}) = |c_i|^2$$

follows within that framework.

At each bounded-definability stage, these fibers carry a finite ledger of relative infinitesimal distinctions. The central structural result is a product-lift theorem: for a normalizable Dirac carrier density $\rho = \psi^\dagger \psi$, every admissible positive, local, translation-covariant, fiber-symmetric lift to that ledger is forced to have the product form

$$\mathcal{L}(\rho) = \rho \, d^3x \otimes \mu.$$

The microscopic fiber measure μ may weight distinct infinitesimal scales arbitrarily, yet after projection to real space that entire freedom survives only as one positive scalar and cancels from every normalized carrier ratio. For alternatives resolved by disjoint spatial supports or orthogonal internal projections, the projective carrier shares are therefore exactly $|c_i|^2$, before any probabilistic assumption is made.

Probability enters only afterward. A common-history selection theorem proves that any branch-blind competing-event process inherits those carrier shares as its conditional outcome probabilities, despite arbitrary common depletion, memory, or colored envi-

ronmental history. Long-run frequencies follow separately under explicit repeated-trial assumptions.

The derivation is projective: it fixes relative outcome probabilities within the Plenum framework rather than an absolute event rate. A conditional effective model illustrates how the derived shares can drive one exclusive local crossing and initiate an irreversible record, but that realization is not a premise of the Born theorem.

Generative AI and AI-assisted technologies were used under the author's direction to assist with drafting, editing, organization, exposition, and presentation. The author originated the research questions, concepts, mathematical constructions, theorem statements, arguments, interpretations, and conclusions; selected the research directions and source materials; independently evaluated, revised, and approved all AI-assisted material; and takes sole responsibility for every statement and result. No AI system is an author, cited authority, or source of mathematical validation.

Keywords: Born rule, quantum measurement problem, infinitesimal structure, non-Archimedean analysis, projective selection, competing risks

1. Introduction

The quantum measurement problem is often first met in a mild form: that to observe a system is to disturb it. Its sharper form is harder to set aside. A quantum system evolves as a superposition, a wave of coexisting alternatives, yet each measurement returns a single definite outcome, and across many measurements those outcomes occur with frequencies fixed exactly by the Born rule. An exact pattern therefore governs which alternative is realized, and how often. The origin of that pattern has remained unknown. Unknown is not unknowable. That a century of work has established the pattern's precision to extraordinary accuracy is not evidence that its source lies beyond reach; it is evidence only that the source had not yet been found.

This work takes up the question with a mathematical instrument that has not, to my knowledge, previously been directed at it. The Plenum Field is an ordered field built as an iterated tower of ultrapowers, in which the familiar real line appears as a quotient: each real number is the shadow of a richer structure, and beneath every real point lies a fiber of distinctions that the passage to the reals erases. These erased distinctions, ordinarily invisible, are the material this paper works with. The starting step was deliberately concrete. I set out to picture how the underlying waves interact, and to ask what happens in the vanishing

structure where a wave approaches zero. Seen from there, the Born pattern need not be taken as primitive. It can be derived.

There is a reason to expect the Plenum to have something to say here, and it is worth stating plainly before any machinery appears. The Born rule is a statement about ratios: it fixes the relative frequencies of outcomes, and nothing more; no absolute scale attaches to it. The Plenum is built the same way. In passing from its richer structure to the ordinary real line, an absolute layer is discarded and only the relationships survive; the reals are what remains after the quotient. A rule that speaks only of ratios and a number system built by discarding all but the ratios are objects of the same kind. That correspondence is not a proof, and I do not offer it as one. But it is the reason the derivation that follows comes out the way it does, and it is why this particular instrument was able to reach the result.

What this paper proves is a single, exact statement. Given the Plenum structure and the ordinary conserved current of the measured particle, the probability that a measurement yields outcome i , conditioned on its yielding any outcome at all, is $|c_i|^2$. The argument runs in a few clean steps: the fibers beneath each real point carry a finite ledger of distinctions; that ledger lifts to a carrier measure whose form is forced, not chosen; and the freedom that remains cancels, leaving only the ratios the Born rule requires. What the paper does not do is equally important, and I will be exact about it throughout. The result holds within the Plenum framework, and it fixes ratios rather than any absolute quantity, which is precisely the content of the Born rule and no less. A physical mechanism realizing the measurement event is offered later, but it is illustrative and conditional; the theorem does not rest on it, and a reader persuaded by the theorem need read no further.

With the intuition in place, we turn to the construction itself.

2. Scope, Status, and What Is Not Claimed

The paper contains two epistemically distinct contributions. Part II proves a framework-relative projective selection theorem. Part III presents one conditional effective realization of a local measurement event. The two contributions are related, but the mathematical proof does not depend on the effective realization.

The Born theorem proved in Part II does not depend on the conditional effective-model realization presented in Part III.

2.1. Framework-relative result

No derivation in quantum foundations proceeds without mathematical and physical premises. The claim here is conditional and exact: given the Plenum quotient structure, the bounded infinitesimal ledger, the ordinary normalizable Dirac carrier, the stated admissibility conditions on its lift, a locally resolved branch decomposition, and a branch-blind first-selection interface, the conditional single-event law follows.

The result can also be read as a structural sufficiency theorem. A quotient fiber carrying bounded positive distinctions, together with a local translation-covariant lift of a conserved L^1 carrier and a branch-blind first-selection interface, is sufficient to produce the Born branch-selection law projectively. The Plenum supplies the concrete realization studied in this paper.

2.2. Projective result

The theorem fixes a measure ray rather than an absolute microscopic carrier normalization:

$$[p_*\mathcal{L}(\rho)] = [\rho d^3x] . \quad (2.1)$$

Absolute fiber mass is neither derived nor required. The Born rule itself fixes relative outcome weights, and the projective theorem supplies exactly that information.

2.3. Single-click law and repeated-trial frequencies

The principal theorem is

$$P(i \mid \text{click}) = |c_i|^2 . \quad (2.2)$$

This is a conditional law for one first marked measurement event. A long-run frequency statement requires additional assumptions concerning repeated preparation, independence, and identical response conditions. The frequency corollary is stated separately and is not used to prove the single-event theorem.

2.4. *Conditional local realization*

Part III asks whether the abstract branch-blind first-selection interface can be realized by a concrete local effective model. The result is illustrative and existence-supported. It shows that one compatible regime can produce a local crossing, rapid exclusion of competitors, a finite compound transition, leakage, and stable local record initiation. It does not establish that every detector uses this mechanism.

2.5. *Empirical scope*

The present paper does not claim a new experimentally distinguishable deviation from standard quantum statistics. Its principal contribution is a derivation of the established Born law within a specified framework. Any future empirical differentiation must arise from a detector-specific calibration of the conditional dynamical layer.

2.6. *Status map*

Established in prior work:

- the Plenum ordered-field construction;
- the real-line quotient and infinitesimal kernel;
- the spatial quotient fibers;
- bounded definability within the kernel;
- the finite-stage Abyss construction.

Proved in this paper:

- whole-fiber canonicity;
- the positivity obstruction for signed displacement;
- the bounded relative-distinction ledger;
- canonical fiber transport;
- the L^1 product-lift theorem;
- the broad finite-Borel counterexample;
- projective cancellation;
- spatial and orthogonal-modal carrier shares;
- the common-history first-selection theorem;

- the framework-relative single-click Born law.

Conditionally realized:

- derived regional edge flow;
- one local crossing;
- binary exclusivity;
- rapid lockout;
- a compound transition;
- environmental leakage;
- stable local record initiation.

Part II. Projective Born Selection

3. Mathematical Inputs and Citation Boundary

This part derives the projective Born selection law from a small set of previously established Plenum results and the ordinary conserved Dirac carrier. The purpose of this section is to state those inputs exactly, fix notation, and separate them from the new results proved below.

3.1. Quotient structure

Let $\mathcal{F}$ denote the Plenum Field, let $\mathcal{F}_{fin}$ be its ordered ring of finite elements, and let

$$I(\mathcal{F}) = \left\{ x \in \mathcal{F} : |x| < \frac{1}{n} \text{ for every positive standard integer } n \right\} \quad (3.1)$$

be its infinitesimal ideal. Prior work establishes that the standard-part map

$$st : \mathcal{F}_{fin} \rightarrow \mathbb{R} \quad (3.2)$$

is a surjective ordered-ring homomorphism with kernel $I(\mathcal{F})$. Consequently,

$$\mathbb{R} \cong \mathcal{F}_{fin}/I(\mathcal{F}). \quad (3.3)$$

Thus two finite Plenum elements have the same real shadow precisely when their difference is infinitesimal. This quotient structure, rather than any physical interpretation of the kernel, is the first inherited input [1, 2].

Define Plenum Space by

$$\mathcal{S} = \mathcal{F}^3, \quad (3.4)$$

and let $\mathcal{S}_{fin}$ denote the componentwise finite part. The componentwise standard-part map is

$$\pi = st_3 : \mathcal{S}_{fin} \rightarrow \mathbb{R}^3. \quad (3.5)$$

Its kernel is

$$K = \ker \pi = (I(\mathcal{F}))^3, \quad (3.6)$$

and therefore

$$\mathbb{R}^3 \cong \mathcal{S}_{fin}/K. \quad (3.7)$$

For each $x \in \mathbb{R}^3$, every choice of $X \in \mathcal{S}_{fin}$ satisfying $\pi(X) = x$ determines the same quotient fiber

$$\mathcal{K}_x = \pi^{-1}(x) = X + K. \quad (3.8)$$

The fiber is therefore an affine coset of the spatial infinitesimal kernel. It is not supplied with a preferred origin.

3.2. Bounded definability inside the kernel

Fix an admissible finite-symbol language L for the expanded Plenum structure and a symbol budget B . Let $E_{L,B}$ denote the set of positive infinitesimals uniquely definable in L by formulas of length at most B . Prior work proves that $E_{L,B}$ is finite. Subject to the stated nonemptiness hypotheses, it therefore has a minimum

$$\Gamma_{L,B} = \min E_{L,B}. \quad (3.9)$$

The same bounded language determines a finite set of uniquely definable positive infinite elements. If $\Lambda_{L,B}$ is its maximum, then

$$A_{L,B} = \frac{\Gamma_{L,B}}{\Lambda_{L,B}} \quad (3.10)$$

is a positive infinitesimal lying below every member of $E_{L,B}$, with $A_{L,B}/\epsilon$ infinitesimal for every $\epsilon \in E_{L,B}$. Iterating the construction through controlled language expansion produces a strictly descending finite-stage tower of named infinitesimal scales. These results are relative to the chosen language, budget, coding convention, and named constants. No absolute smallest infinitesimal and no presentation-independent global scale are asserted [3].

Only two consequences of that prior construction are required in the present proof:

1. at each fixed (L, B) , the set $E_{L,B}$ is finite;
2. every $\epsilon \in E_{L,B}$ lies in $I(\mathcal{F})$, so every signed coordinate triple formed from ϵ lies in K .

The full Abyss construction and tower are not reproved here.

3.3. Citation boundary

The prior Plenum papers establish the field construction, the real-line quotient, the spatial kernel fibers, bounded definability, and the finite-stage Abyss construction. They do not establish the canonical-fiber theorem, the relative-distinction ledger, the product-lift theorem, projective cancellation, or the Born selection law [1, 2, 3].

Those results begin in the next section and are new to the present paper. The distinction will be maintained throughout:

$$\begin{array}{ll} \text{Inherited:} & \mathcal{F}, I(\mathcal{F}), \pi, K, E_{L,B}, \text{ and the bounded tower;} \\ \text{proved here:} & \text{canonical fibers, relative ledgers, and product lifts,} \\ & \text{projective carrier uniqueness, and Born selection.} \end{array} \quad (3.11)$$

No physical measure, probability law, or measurement dynamics has been introduced at this stage.

4. Canonical Fibers and Bounded Relative Distinctions

The quotient construction supplies an entire kernel fiber beneath each point of real space, but it does not select a preferred representative within that fiber. The first task is therefore to determine what sub-shadow object is canonical under the natural vertical symmetry. The second is to identify what kind of positive ledger can be built from the infinitesimal distinctions retained by the fiber.

4.1. Whole-fiber canonicity

The additive group K acts on each fiber $\mathcal{K}_x$ by vertical translation:

$$\tau_\kappa(Y) = Y + \kappa, \quad \kappa \in K, \quad Y \in \mathcal{K}_x. \quad (4.1)$$

Theorem 4.1. Whole-Fiber Canonicity

Let $x \in \mathbb{R}^3$, and let $\mathcal{K}_x = \pi^{-1}(x)$. The vertical action of K on $\mathcal{K}_x$ is simply transitive. Consequently, if $A \subseteq \mathcal{K}_x$ is nonempty and satisfies

$$A + \kappa = A \quad \text{for every } \kappa \in K, \quad (4.2)$$

then

$$A = \mathcal{K}_x. \quad (4.3)$$

Proof. First, the action is free. If $Y + \kappa = Y$, then cancellation in the additive group of S gives $\kappa = 0$.

It is also transitive. Let $Y, Z \in \mathcal{K}_x$. Since

$$\pi(Z - Y) = \pi(Z) - \pi(Y) = x - x = 0, \quad (4.4)$$

one has $Z - Y \in K$. Hence $Z = Y + (Z - Y)$, so a vertical kernel translation carries Y to Z .

Now let A be nonempty and invariant under every vertical translation. Choose $Y \in A$. For arbitrary $Z \in \mathcal{K}_x$, the preceding argument gives $Z - Y \in K$. By invariance, $Z = Y + (Z - Y) \in A$. Thus every point of $\mathcal{K}_x$ lies in A , and therefore $A = \mathcal{K}_x$. $\square$

The meaning of “canonical” in Theorem 4.1 is exact and limited: canonical means invariant under the full vertical kernel action. The theorem does not deny that a representative or proper subset can be selected after additional structure is supplied. It shows that no such selection is determined by the quotient fiber alone.

4.2. Positivity obstruction

A kernel displacement is signed. It therefore cannot itself serve as a nonzero positive additive content.

Lemma 4.2. Positive-Ledger Obstruction

Let

$$q : K \rightarrow \mathbb{R}_{\geq 0} \tag{4.5}$$

be additive. Then q is identically zero.

Proof. For every $\kappa \in K$,

$$q(\kappa) + q(-\kappa) = q(0) = 0. \tag{4.6}$$

Both terms are nonnegative, so both vanish. In particular, $q(\kappa) = 0$ for every $\kappa \in K$. $\square$

The lemma rules out a positive ledger obtained by summing signed displacement. It does not force a uniform counting measure. It leaves open the more general possibility of assigning positive weights to distinct displacement types. That distinction will be essential later, since the available symmetries do not force equal weights across different infinitesimal scales.

4.3. The bounded relative-distinction ledger

For each $\epsilon \in E_{L,B}$, define the signed coordinate orbit

$$C_\epsilon = \{(\sigma_1\epsilon, \sigma_2\epsilon, \sigma_3\epsilon) : \sigma_i \in \{-1, +1\}\} \subseteq K. \quad (4.7)$$

The bounded relative-distinction ledger is

$$D_{L,B} = \bigcup_{\epsilon \in E_{L,B}} C_\epsilon. \quad (4.8)$$

The elements of $D_{L,B}$ are relative kernel displacements. They are not absolute points chosen inside a fiber.

Theorem 4.3. Bounded Relative-Ledger Theorem

At every fixed bounded-definability stage (L, B) , the set $D_{L,B}$ is a finite, basepoint-independent ledger of relative kernel displacement types.

More precisely:

1. $D_{L,B} \subseteq K$;
2. $D_{L,B}$ is finite;
3. each C_ϵ is an orbit under the natural signed-coordinate and coordinate-permutation symmetries of the additive spatial kernel;
4. no preferred representative of a quotient fiber is required to define $D_{L,B}$.

Proof. Let $d = (\sigma_1\epsilon, \sigma_2\epsilon, \sigma_3\epsilon) \in D_{L,B}$. Since $\epsilon \in I(\mathcal{F})$, each coordinate of d is infinitesimal. Therefore $d \in K$.

The set $E_{L,B}$ is finite, and each C_ϵ contains eight signed triples. Hence $D_{L,B}$ is finite.

The group

$$G_{sc} = \{-1, +1\}^3 \rtimes S_3 \quad (4.9)$$

acts on K by independent coordinate sign changes and coordinate permutations. This is an additive spatial symmetry; no claim is made that every such transformation is an automorphism of the ambient ordered field. For fixed ϵ , the

action is transitive on C_ϵ . No symmetry is assumed to exchange C_ϵ with $C_{\epsilon'}$ when $\epsilon \neq \epsilon'$.

Finally, $D_{L,B}$ is defined as a set of displacement vectors in K . If $Y \in \mathcal{K}_x$ is used to realize the ledger inside the fiber, the corresponding point set is $Y + D_{L,B}$. Replacing Y by $Y + \kappa_0$, with $\kappa_0 \in K$, replaces this realization by

$$Y + \kappa_0 + D_{L,B} = (Y + D_{L,B}) + \kappa_0. \quad (4.10)$$

The absolute placement changes, but the relative displacement labels $D_{L,B}$ do not. The ledger is therefore basepoint-independent in the required sense. $\square$

Remark 4.1. Weighting freedom

The symmetry group in (4.9) forces equal treatment only within a fixed orbit C_ϵ . It does not force equal treatment of distinct scale orbits. A general positive symmetric fiber measure may therefore take the form

$$\mu = \sum_{\epsilon \in E_{L,B}} a_\epsilon \#_{C_\epsilon}, \quad a_\epsilon \geq 0, \quad (4.11)$$

where $\#_{C_\epsilon}$ is counting measure on C_ϵ . Uniform microscopic counting corresponds to the special choice $a_\epsilon = 1$, but it is neither assumed nor derived. The later projective theorem will show that the freedom in the coefficients a_ϵ does not affect branch ratios.

4.4. Canonical transport between fibers

For $v \in \mathcal{S}_{fin}$, define

$$T_v(Y) = Y + v. \quad (4.12)$$

Theorem 4.4. Canonical Fiber Transport

For every $x \in \mathbb{R}^3$ and every $v \in \mathcal{S}_{fin}$, the map T_v restricts to a bijection

$$T_v : \mathcal{K}_x \rightarrow \mathcal{K}_{x+\pi(v)}. \quad (4.13)$$

Moreover, it preserves every relative kernel displacement:

$$T_v(Y + \kappa) - T_v(Y) = \kappa \quad (4.14)$$

for all $Y \in \mathcal{K}_x$ and $\kappa \in K$. If $v, v' \in \mathcal{S}_{fin}$ have the same real shadow, then the two transports differ only by a vertical kernel translation.

Proof. If $Y \in \mathcal{K}_x$, then

$$\pi(T_v(Y)) = \pi(Y + v) = \pi(Y) + \pi(v) = x + \pi(v). \quad (4.15)$$

Thus $T_v(Y) \in \mathcal{K}_{x+\pi(v)}$. The inverse map is T_{-v} , so the restriction is bijective.

For $\kappa \in K$,

$$T_v(Y + \kappa) - T_v(Y) = (Y + \kappa + v) - (Y + v) = \kappa. \quad (4.16)$$

Finally, if $\pi(v) = \pi(v')$, then $\pi(v' - v) = 0$, so $v' - v \in K$. Therefore $T_{v'}(Y) = T_v(Y) + (v' - v)$, and the two transports differ only by a vertical kernel translation. $\square$

Corollary 4.5. Fiberwise Constancy of the Bounded Ledger

At fixed (L, B) , every quotient fiber carries the same relative ledger type $D_{L,B}$, and every finite Plenum translation preserves that type.

4.5. Checkpoint

The argument has now established four facts:

- each real point corresponds canonically to an entire kernel fiber;
- signed displacement cannot itself be positive additive content;
- each bounded stage supplies a finite relative-distinction ledger;
- the ledger is transported identically between quotient fibers.

No probability interpretation has been assigned to the Dirac density. No physical event rate, collapse law, or Born coefficient has been introduced. The next section asks a purely structural question: given a normalizable L^1 carrier

density and the finite ledger derived above, what form can a positive, local, translation-covariant lift take?

5. The L^1 Product-Lift Theorem

The preceding section established that each bounded-definability stage supplies the same finite relative-distinction ledger over every real location. It did not assign a measure to that ledger.

This section introduces the conserved carrier and asks a constrained structural question. Given a normalizable carrier density on real space, what positive measure lifts are compatible with locality, spatial translation covariance, and the symmetries of the bounded ledger?

The admissible class is defined first. Product form and projective cancellation are not included among its defining properties.

5.1. The physical carrier domain

Let $X = \mathbb{R}^3$ with Lebesgue measure dx . At a fixed time, let ψ be a normalizable Dirac state,

$$\psi \in L^2(X, \mathbb{C}^4), \quad (5.1)$$

and define its carrier density by

$$\rho_e(x) = j_e^0(x) = \psi^\dagger(x)\psi(x). \quad (5.2)$$

Then

$$\rho_e \in L_+^1(X), \quad \|\rho_e\|_1 = \|\psi\|_2^2. \quad (5.3)$$

No probability interpretation is assigned to ρ_e at this stage. It enters only as the nonnegative density of the conserved Dirac carrier.

The theorem below is therefore formulated on

$$L_+^1(X) = \{\rho \in L^1(X) : \rho \geq 0 \text{ almost everywhere}\}. \quad (5.4)$$

This domain is fixed by the form of a normalizable Dirac carrier. It is not selected to obtain the desired branch ratios.

Fix one bounded-definability stage (L, B) , and abbreviate its finite ledger by $D = D_{L,B}$. Let G_D denote the finite group generated by the signed-coordinate and coordinate-permutation symmetries acting within the orbits $C_\epsilon \subseteq D$.

For $\rho \in L_+^1(X)$, write

$$\nu_\rho(A) = \int_A \rho(x) dx \quad (5.5)$$

for its associated finite carrier measure.

5.2. Admissible lifts

A ledger lift is a map

$$\mathcal{L} : L_+^1(X) \rightarrow \mathcal{M}_+(X \times D), \quad (5.6)$$

where $\mathcal{M}_+(X \times D)$ is the cone of finite positive Borel measures on $X \times D$.

Definition 5.1. Admissible Ledger Lift

The lift $\mathcal{L}$ is admissible if it satisfies the following conditions.

A1. Additivity and positive homogeneity. For all $\rho_1, \rho_2 \in L_+^1(X)$ and $a, b \geq 0$,

$$\mathcal{L}(a\rho_1 + b\rho_2) = a\mathcal{L}(\rho_1) + b\mathcal{L}(\rho_2). \quad (5.7)$$

A2. Base locality. For every Borel set $A \subseteq X$,

$$\rho = 0 \text{ almost everywhere on } A \Rightarrow \mathcal{L}(\rho)(A \times D) = 0. \quad (5.8)$$

A3. Spatial translation covariance. For $v \in X$, let $\tau_v(x) = x + v$ and $(\tau_v\rho)(x) = \rho(x - v)$. The lift satisfies

$$\mathcal{L}(\tau_v\rho) = (\tau_v \times id_D)_* \mathcal{L}(\rho). \quad (5.9)$$

A4. Fiber-symmetry invariance. For every $g \in G_D$,

$$(id_X \times g)_* \mathcal{L}(\rho) = \mathcal{L}(\rho). \quad (5.10)$$

No symmetry is assumed to exchange distinct scale orbits C_ϵ and $C_{\epsilon'}$.

A5. Finite nontrivial output. For every nonzero $\rho \in L_+^1(X)$,

$$0 < \mathcal{L}(\rho)(X \times D) < \infty. \quad (5.11)$$

These conditions do not mention Born coefficients, branch amplitudes, projective cancellation, equal microscopic counting, or any proposed probability law.

5.3. Slice operators

For each $d \in D$, define

$$\mathcal{L}_d(\rho)(A) = \mathcal{L}(\rho)(A \times \{d\}). \quad (5.12)$$

Base locality implies

$$\mathcal{L}_d(\rho) \ll \nu_\rho \ll dx. \quad (5.13)$$

By the Radon-Nikodym theorem, there is a unique $T_d \rho \in L_+^1(X)$, up to equality almost everywhere, such that

$$\mathcal{L}_d(\rho)(A) = \int_A (T_d \rho)(x) dx. \quad (5.14)$$

Lemma 5.2. Positive Linear Extension

For each $d \in D$, the map $T_d : L_+^1(X) \rightarrow L_+^1(X)$ extends uniquely to an everywhere-defined positive linear operator

$$T_d : L^1(X) \rightarrow L^1(X). \quad (5.15)$$

The extended operator is local:

$$f = 0 \text{ almost everywhere on } A \Rightarrow T_d f = 0 \text{ almost everywhere on } A. \quad (5.16)$$

Proof. Additivity and positive homogeneity of $\mathcal{L}$ pass to every slice measure and hence to its Radon-Nikodym density. For $f \in L^1(X)$, choose any decomposition $f = u - v$ with $u, v \in L_+^1(X)$ and define $T_d f = T_d u - T_d v$. If also $f = u' - v'$, then $u + v' = u' + v$, so cone additivity gives $T_d u + T_d v' = T_d u' + T_d v$ and the extension is well defined. Positivity and linearity follow directly. If $f = 0$ almost everywhere on A , both f_+ and f_- vanish there, and base locality applied separately gives the locality conclusion. $\square$

5.4. Automatic boundedness

No continuity or boundedness hypothesis was included in Definition 5.1. Boundedness follows from positivity and the requirement that the lift be finite for every L^1 carrier.

Lemma 5.3. Automatic Boundedness

Every slice operator $T_d : L^1(X) \rightarrow L^1(X)$ constructed above is bounded.

Proof. A positive linear operator satisfies

$$|T_d f| \leq T_d |f| \quad (5.17)$$

almost everywhere. Suppose, for contradiction, that T_d is unbounded. For every positive integer n , choose $g_n \in L_+^1(X)$ with $\|g_n\|_1 = 1$ and $\|T_d g_n\|_1 \geq 2^n$. Set $\rho_n = 2^{-n} g_n$. Then $\|\rho_n\|_1 = 2^{-n}$ and $\|T_d \rho_n\|_1 \geq 1$.

The series $\rho = \sum_{n=1}^{\infty} \rho_n$ converges in $L_+^1(X)$. For every N , positivity and finite linearity imply

$$T_d \rho \geq \sum_{n=1}^N T_d \rho_n. \quad (5.18)$$

All functions on the right are nonnegative. Consequently,

$$\|T_d \rho\|_1 \geq \sum_{n=1}^N \|T_d \rho_n\|_1 \geq N. \quad (5.19)$$

This holds for every N , contradicting $T_d \rho \in L^1(X)$. Thus T_d is bounded. $\square$

The argument uses no countable additivity of the lift rule as an operator-valued map. It uses only the existence of $T_d\rho$ for one L^1 function, finite linearity, positivity, and finiteness.

5.5. Translation covariance and the centralizer identity

Translation covariance passes to each slice:

$$T_d(\tau_v f) = \tau_v(T_d f). \quad (5.20)$$

Let convolution be

$$(f * h)(x) = \int_X f(y)h(x - y) dy. \quad (5.21)$$

Lemma 5.4. Centralizer Identity

For all $f, h \in L^1(X)$,

$$T_d(f * h) = f * T_d h. \quad (5.22)$$

Proof. For fixed h , the convolution is the L^1 -valued Bochner integral

$$f * h = \int_X f(y)\tau_y h dy. \quad (5.23)$$

Since T_d is bounded, it may be passed through the integral. Translation covariance then gives

$$T_d(f * h) = \int_X f(y)\tau_y (T_d h) dy = f * T_d h. \quad (5.24)$$

□

5.6. Wendel representation

The additive group $G = (\mathbb{R}^3, +)$ is locally compact, and Lebesgue measure is its Haar measure. Wendel's centralizer theorem applies to a bounded linear operator $A : L^1(G) \rightarrow L^1(G)$ satisfying

$$A(f * h) = f * A(h). \quad (5.25)$$

It yields a unique regular Borel measure η of bounded variation such that $Af = f * \eta$ and $\|A\| = |\eta|(G)$.

Applying the theorem to T_d gives a unique finite regular measure of bounded variation η_d satisfying

$$T_d f = f * \eta_d. \quad (5.26)$$

The representation theorem used here is Wendel's centralizer theorem [4, Theorem 1].

Lemma 5.5. Positivity of the Representing Measure

The representing measure η_d is positive.

Proof. Let (e_α) be a nonnegative L^1 approximate identity with integral one. Since T_d is positive, $e_\alpha * \eta_d = T_d e_\alpha \geq 0$. Testing against nonnegative $\varphi \in C_c(X)$, applying Fubini, and taking the approximate-identity limit by dominated convergence with respect to $|\eta_d|$ yields

$$\int_X \varphi d\eta_d \geq 0. \quad (5.27)$$

Hence η_d is positive. □

5.7. Base locality collapses the convolution kernel

Lemma 5.6. Locality-to-Point-Mass Lemma

If $T_d f = f * \eta_d$ is base-local, then

$$\text{supp} \eta_d \subseteq \{0\}. \quad (5.28)$$

Consequently, there is a constant $c_d \geq 0$ such that

$$\eta_d = c_d \delta_0, \quad T_d f = c_d f. \quad (5.29)$$

Proof. Suppose that $y_0 \neq 0$ lies in the support of η_d . Choose $r > 0$ sufficiently small that $B(0, 2r)$ and $B(y_0, r/2)$ are disjoint. Let $f = \mathbf{1}_{B(0, 2r)}$ and $A = B(y_0, r/2)$. Then $f = 0$ on A , but for $x, y \in A$ one has $x - y \in B(0, 2r)$, so

$$(T_d f)(x) \geq \eta_d(A) > 0 \quad (5.30)$$

for every $x \in A$. This contradicts locality. Therefore the support is contained in $\{0\}$, and a positive finite measure on that singleton is $c_d \delta_0$. $\square$

5.8. *Product-lift uniqueness*

Theorem 5.7. L^1 Product-Lift Theorem

Let $D = D_{L,B}$ be a finite bounded relative-distinction ledger. Every admissible lift

$$\mathcal{L} : L_+^1(X) \rightarrow \mathcal{M}_+(X \times D) \quad (5.31)$$

has the form

$$\boxed{\mathcal{L}(\rho) = \rho(x) dx \otimes \mu} \quad (5.32)$$

for one fixed finite positive measure μ on D , independent of ρ and of the spatial point x .

The measure μ is invariant under G_D . Its weights are equal within each orbit C_ϵ , but need not be equal across distinct scale orbits.

Proof. For each $d \in D$, Lemmas 5.2 through 5.6 give a constant $c_d \geq 0$ such that

$$\mathcal{L}_d(\rho)(A) = c_d \int_A \rho(x) dx. \quad (5.33)$$

Define $\mu = \sum_{d \in D} c_d \delta_d$. Since D is finite, μ is finite and positive. Equality on the generating rectangles $A \times B$ gives $\mathcal{L}(\rho) = \nu_\rho \otimes \mu$. Fiber-symmetry invariance gives $c_{g(d)} = c_d$. Finally, finite nontrivial output implies $0 < \mu(D) < \infty$. $\square$

Corollary 5.8. Uniqueness of the Fiber Factor

For a fixed admissible lift $\mathcal{L}$, the measure μ in (5.32) is unique.

5.9. Assumption ledger

Theorem 5.7 assumes a normalizable L^1 carrier; positivity, additivity, and positive homogeneity; base locality; spatial translation covariance; fiber-symmetry invariance; and finite nontrivial output.

It does not assume product form, projective cancellation, Born weights, operator continuity or boundedness, barycentricity, normality, module-linearity, or uniform microscopic counting. Boundedness is proved in Lemma 5.3. Module-linearity is obtained as the centralizer identity in Lemma 5.4. Product form is the conclusion of Theorem 5.7.

6. Audit Note: Two Failed Extensions

The product-lift theorem was not reached by assuming that every plausible carrier lift must have product form. Two broader routes were examined and rejected because they permit precisely the kind of hidden carrier assumption that the final theorem is intended to exclude.

6.1. Failure on the domain of all finite Borel measures

Let $\mathcal{M}_+(X)$ denote the cone of finite positive Borel measures on X . Every $\nu \in \mathcal{M}_+(X)$ has a unique decomposition

$$\nu = \nu_{at} + \nu_{na}, \quad (6.1)$$

where ν_{at} is atomic and ν_{na} is nonatomic.

Choose two finite positive G_D -invariant measures μ_0 and μ_1 on D , with positive total mass, and define

$$\mathcal{L}_B(\nu) = \nu \otimes \mu_0 + \nu_{na} \otimes \mu_1. \quad (6.2)$$

Proposition 6.1. Counterexample to the Broad Measure-Domain Theorem

The lift $\mathcal{L}_B$ satisfies the natural analogues of positivity, additivity, positive homogeneity, base locality, spatial translation covariance, fiber-symmetry invariance, and finite nontrivial output on $\mathcal{M}_+(X)$. Nevertheless, no fixed measure μ on D satisfies

$$\mathcal{L}_B(\nu) = \nu \otimes \mu \quad (6.3)$$

for every finite positive Borel measure ν .

Proof. For a Dirac measure δ_x , the nonatomic part vanishes, so $\mathcal{L}_B(\delta_x) = \delta_x \otimes \mu_0$. For a nonzero nonatomic measure λ , one has $\mathcal{L}_B(\lambda) = \lambda \otimes (\mu_0 + \mu_1)$. Since $\mu_1 \neq 0$, no single fixed fiber measure gives both outputs. $\square$

The pushforward also fails to be a fixed scalar multiple of mixed atomic and nonatomic inputs:

$$p_*\mathcal{L}_B(\nu) = \mu_0(D)\nu + \mu_1(D)\nu_{na}. \quad (6.4)$$

6.2. Why the counterexample does not apply to the Dirac-density domain

The physical carrier used in Theorem 5.7 has the form $\nu_\rho = \rho dx$, with $\rho \in L^1_+(X)$. Every such measure is nonatomic. Consequently, the counterexample collapses to

$$\mathcal{L}_B(\nu_\rho) = \nu_\rho \otimes (\mu_0 + \mu_1), \quad (6.5)$$

which is again a product lift.

The failed theorem shows that the L^1 carrier restriction matters. The final theorem returns to the physical carrier domain and proves product form there

from positive-operator theory, translation covariance, and locality. The domain is supplied by the carrier rather than selected to force the answer.

6.3. *Rejected matrix-carrier route*

A separate proposed route attempted to obtain fixed carrier content from a conserved matrix trace. Such a route would require independent derivations that the relevant sub-shadow distinctions are necessarily represented by matrix degrees of freedom, that spectral or matrix locality implies the spatial or modal locality required for separated alternatives, and that the resulting conserved matrix current is identified with or transported comovingly with the conserved Dirac carrier.

Those identifications were not supplied by the inherited Plenum construction. Trace conservation alone therefore could not establish the physical carrier relation required by the Born argument. The matrix-carrier route is not used in this paper.

7. Projective Cancellation

The L^1 product-lift theorem determines the spatial dependence of every admissible lift but leaves the finite fiber measure μ undetermined across distinct scale orbits. This section shows that the remaining freedom contributes only one positive global scalar.

7.1. *Measure rays*

Definition 7.1. Measure Rays

For nonzero finite positive measures ν and ν' , write $\nu \sim \nu'$ if $\nu' = a\nu$ for some $a > 0$. The equivalence class $[\nu]$ is the measure ray determined by ν .

7.2. *Pushforward of the product lift*

Let $p : X \times D \rightarrow X$ be $p(x, d) = x$.

Theorem 7.2. Projective Pushforward Theorem

If $\mathcal{L}(\rho) = \nu_\rho \otimes \mu$, then

$$p_*\mathcal{L}(\rho) = \mu(D)\nu_\rho. \tag{7.1}$$

For every nonzero $\rho \in L_+^1(X)$,

$$\boxed{[p_*\mathcal{L}(\rho)] = [\nu_\rho]} . \quad (7.2)$$

Proof. For every Borel set A ,

$$p_*\mathcal{L}(\rho)(A) = \mathcal{L}(\rho)(A \times D) = \nu_\rho(A)\mu(D). \quad (7.3)$$

Since $0 < \mu(D) < \infty$, the two measures determine the same ray. $\square$

7.3. Full cancellation of cross-scale weighting freedom

By Remark 4.1 and Theorem 5.7,

$$\mu = \sum_{\epsilon \in E_{L,B}} a_\epsilon \#_{C_\epsilon}, \quad a_\epsilon \geq 0. \quad (7.4)$$

Because each ϵ is strictly positive, the eight sign vectors produce eight distinct elements of C_ϵ . Hence

$$\mu(D) = 8 \sum_{\epsilon \in E_{L,B}} a_\epsilon. \quad (7.5)$$

All microscopic weighting freedom is contained in this one scalar.

Corollary 7.3. Regional-Ratio Invariance

For Borel sets $A, B \subseteq X$ with $\nu_\rho(B) > 0$,

$$\frac{p_*\mathcal{L}(\rho)(A)}{p_*\mathcal{L}(\rho)(B)} = \frac{\nu_\rho(A)}{\nu_\rho(B)}. \quad (7.6)$$

For a finite measurable partition or projected decomposition $\{U_i\}$,

$$\frac{p_*\mathcal{L}(\rho)(U_i)}{\sum_j p_*\mathcal{L}(\rho)(U_j)} = \frac{\nu_\rho(U_i)}{\sum_j \nu_\rho(U_j)}. \quad (7.7)$$

The coefficients a_ϵ may be unequal, may vanish on some scale orbits, and may vary by bounded stage. Provided the total fiber measure is positive and finite, none affects a normalized regional ratio. Uniform microscopic counting is neither needed nor implied.

7.4. Finite-stage stability

Let D_n be the finite ledger at stage n , and let $\mathcal{L}_n(\rho) = \nu_\rho \otimes \mu_n$ with $0 < \mu_n(D_n) < \infty$.

Theorem 7.4. Finite-Stage Projective Constancy

For every finite stage n ,

$$p_{n*}\mathcal{L}_n(\rho) = \mu_n(D_n) \nu_\rho, \quad [p_{n*}\mathcal{L}_n(\rho)] = [\nu_\rho]. \quad (7.8)$$

The projective carriers remain constant even if $\mu_n(D_n) \rightarrow \infty$. No finite measure on an infinite completed ledger is required.

7.5. What has and has not been proved

The projective theorem establishes

$$[p_*\mathcal{L}(\rho)] = [\rho dx]. \quad (7.9)$$

It does not yet identify regional carrier ratios with probabilities. No branch-selection process has yet been invoked.

8. From Projective Carrier Ratios to the Single-Click Law

The preceding section established a projective statement about carrier measures. It did not assign probabilities to the carrier density or its regional components. This section first proves that alternatives resolved by disjoint spatial support or by orthogonal internal modes inherit the squared amplitudes as their normalized carrier shares. It then applies an independently stated first-selection theorem. Probability enters only in that second step.

8.1. Local outcome projections

Let $\mathcal{V}$ be a finite-dimensional internal multiplicity space. Consider a normalized carrier state

$$\Psi \in L^2(X, \mathbb{C}^4 \otimes \mathcal{V}), \quad \|\Psi\|_2 = 1, \quad (8.1)$$

with total carrier density $\rho_e(x) = \Psi^\dagger(x)\Psi(x)$.

Let $P_1(x), \dots, P_m(x)$ be measurable orthogonal projection fields satisfying, almost everywhere,

$$P_i^2 = P_i, \quad P_i^\dagger = P_i, \quad P_i P_j = 0 \ (i \neq j), \quad \sum_i P_i = I \quad (8.2)$$

on the outcome subspace occupied by Ψ .

Define

$$\rho_{e,i}(x) = \Psi^\dagger(x) P_i(x) \Psi(x). \quad (8.3)$$

Each $\rho_{e,i}$ is nonnegative and belongs to $L_+^1(X)$, and

$$\rho_e(x) = \sum_i \rho_{e,i}(x). \quad (8.4)$$

For spatial separation, take $P_i(x) = \mathbf{1}_{U_i}(x)I$ for pairwise disjoint regions U_i . For orthogonal internal-mode separation, take $P_i = I_4 \otimes Q_i$ with mutually orthogonal projections Q_i on $\mathcal{V}$. The alternatives may then occupy the same spatial region.

The use of local projection densities is essential. Global Hilbert-space orthogonality alone makes integrated cross terms vanish, but need not produce the pointwise positive decomposition required by the carrier lift.

8.2. Projected branch decomposition

Assume

$$\Psi = \sum_{i=1}^m c_i \Phi_i, \quad \|\Phi_i\|_2 = 1, \quad \sum_i |c_i|^2 = 1, \quad (8.5)$$

and

$$P_i \Phi_j = \delta_{ij} \Phi_j \quad (8.6)$$

almost everywhere.

Lemma 8.1. Local Projected-Density Decomposition

Under these hypotheses,

$$\rho_{e,i}(x) = |c_i|^2 \Phi_i^\dagger(x) \Phi_i(x) \quad (8.7)$$

almost everywhere, and

$$\int_X \rho_{e,i}(x) dx = |c_i|^2. \quad (8.8)$$

Proof. Expand $\Psi^\dagger P_i \Psi$. Because P_i is self-adjoint and $P_i \Phi_j = \delta_{ij} \Phi_j$, every term vanishes except $j = k = i$. Integration then uses $\|\Phi_i\|_2 = 1$. $\square$

Corollary 8.2. Spatially Separated Alternatives

If Φ_i is supported in U_i , with the U_i pairwise disjoint up to null sets, then

$$\int_{U_i} \rho_e(x) dx = |c_i|^2. \quad (8.9)$$

Corollary 8.3. Orthogonal Internal-Mode Alternatives

If $P_i = I_4 \otimes Q_i$ and $(I_4 \otimes Q_i) \Phi_j = \delta_{ij} \Phi_j$, then the alternatives may overlap spatially, but

$$\int_X \Psi^\dagger (I_4 \otimes Q_i) \Psi dx = |c_i|^2. \quad (8.10)$$

No spatial separation is used in this proof. The cross terms vanish because the projections resolve orthogonal internal sectors pointwise.

8.3. Plenum carrier shares

Apply the admissible lift separately to each projected density:

$$\mathcal{L}(\rho_{e,i}) = \rho_{e,i}(x) dx \otimes \mu. \quad (8.11)$$

The same fiber measure appears for every i because Theorem 5.7 assigns one fixed fiber factor to the lift, independently of the input density.

Let $W = \mu(D) > 0$ and define $\nu_{f,i} = p_* \mathcal{L}(\rho_{e,i})$. Then

$$\nu_{f,i} = W \rho_{e,i}(x) dx. \quad (8.12)$$

Let $F_i = \nu_{f,i}(X)$.

Theorem 8.4. Projective Born-Weight Theorem

Under the local projection hypotheses and branch decomposition above,

$$\boxed{\frac{F_i}{\sum_j F_j} = |c_i|^2}. \quad (8.13)$$

The conclusion holds for alternatives separated by disjoint spatial support and for co-located alternatives separated by orthogonal internal modes.

Proof. By Lemma 8.1, $F_i = W |c_i|^2$. The common scalar cancels after normalization. $\square$

Equation (8.13) is still a carrier-ratio statement. The quantities F_i have not yet been declared to be probabilities.

8.4. Conserved edge flow

Proposition 8.5. Conserved Edge-Flow Consequence

Let $\nabla_\mu j_e^\mu = 0$ and $j_e^0 = \rho_e$. At a fixed bounded stage, define

$$J_f^\mu = W j_e^\mu. \quad (8.14)$$

Then $\nabla_\mu J_f^\mu = 0$, and for every sufficiently regular region U ,

$$\frac{d}{dt} \int_U J_f^0 d^3x = - \int_{\partial U} \mathbf{J}_f \cdot \mathbf{n} dS. \quad (8.15)$$

Thus the regional Plenum-associated carrier is fixed-content, comoving with the Dirac carrier, and altered only by boundary flux.

Proof. Because W is constant at the fixed bounded stage, $\nabla_\mu J_f^\mu = W \nabla_\mu j_e^\mu = 0$. Integrating this continuity equation over U and applying the divergence theorem gives the boundary-flux identity above. $\square$

8.5. The first-selection interface

Let

$$w_i = \frac{F_i}{\sum_j F_j} = |c_i|^2. \quad (8.16)$$

Consider a simple competing-risks process with first-event time T and outcome mark $I \in \{1, \dots, m\}$. Let $\mathcal{H}_t$ denote the complete pre-event history. Assume the cause-specific conditional intensities have the form

$$\lambda_i(t \mid \mathcal{H}_t) = w_i \Lambda(t \mid \mathcal{H}_t), \quad (8.17)$$

where Λ is a nonnegative predictable common-history intensity. It may depend arbitrarily on shared depletion, memory, and colored environmental evolution. It introduces no additional branch-dependent weighting beyond the derived carrier shares.

8.6. Common-history selection theorem

Theorem 8.6. Common-History First-Selection Theorem

If

$$\lambda_i(t \mid \mathcal{H}_t) = w_i \Lambda(t \mid \mathcal{H}_t), \quad w_i \geq 0, \quad \sum_i w_i = 1, \quad (8.18)$$

and a first event occurs with positive probability, then

$$\boxed{\mathbb{P}(I = i \mid T < \infty) = w_i.} \quad (8.19)$$

The conclusion is unchanged by arbitrary common non-Markovian history, common depletion, or colored common evolution contained in Λ .

Proof. Let $Y(t) = \mathbf{1}_{\{T \geq t\}}$. The compensator identity gives

$$\mathbb{P}(I = i, T < \infty) = \mathbb{E} \left[\int_0^\infty Y(t) \lambda_i(t \mid \mathcal{H}_t) dt \right]. \quad (8.20)$$

Substitution of (8.18) extracts w_i . Summing over i identifies the remaining expectation with $\mathbb{P}(T < \infty)$. Division yields (8.19). $\square$

This is the standard compensation identity for counting processes with predictable compensators; see Andersen et al. [5, Ch. II].

8.7. Framework-Relative Single-Click Born Selection

Theorem 8.7. Framework-Relative Single-Click Born Selection

Assume:

1. a normalized Dirac carrier admits the locally resolved branch decomposition in (8.5) and (8.6);
2. the alternatives are separated either by disjoint spatial support or by orthogonal internal projections;
3. the Plenum ledger lift satisfies the admissibility hypotheses of Theorem 5.7;
4. the first-selection interface is branch-blind in the common-history sense of (8.17), with w_i the normalized carrier share derived in Theorem 8.4;
5. a first local measurement event occurs with positive probability.

Then

$$\boxed{\mathbb{P}(I = i \mid \text{click}) = |c_i|^2.} \quad (8.21)$$

Proof. Theorem 8.4 gives $w_i = |c_i|^2$. Theorem 8.6 gives $\mathbb{P}(I = i \mid \text{click}) = w_i$. Combining the identities yields (8.21). $\square$

The theorem is framework-relative and projective. It derives conditional branch-selection probabilities, not an absolute event rate or absolute microscopic ledger normalization. The theorem does not depend on the conditional effective realization developed in Part III.

8.8. Repeated-trial corollary

Under independent and identically distributed repeated trials with positive click probability, the strong law of large numbers gives

$$\frac{\sum_{n=1}^N X_{n,i}}{\sum_{n=1}^N Y_n} \rightarrow |c_i|^2 \quad (8.22)$$

almost surely. These repeated-trial assumptions are not part of Theorem 8.7.

9. Why the Derivation Is Not Circular

A derivation of the Born rule is circular if the desired squared-amplitude weights enter through an assumption that already has probabilistic or branch-selective content. The proof above was organized to expose and exclude each plausible entry point.

9.1. The Dirac density is used as a carrier

The density $\rho_e = \Psi^\dagger \Psi$ is used first as a nonnegative conserved carrier. The product-lift and projective-cancellation theorems remain valid for every nonzero $\rho \in L_+^1(X)$, regardless of normalization. The identity

$$\int_X \Psi^\dagger P_i \Psi \, dx = |c_i|^2 \quad (9.1)$$

is established first as a Hilbert-space carrier decomposition. No probability interpretation is used to prove it.

9.2. The microscopic fiber weights are not fitted

The general admissible fiber measure is

$$\mu = \sum_{\epsilon \in E_{L,B}} a_\epsilon \#_{C_\epsilon}, \quad a_\epsilon \geq 0. \quad (9.2)$$

The coefficients may be unequal, zero on some orbits, concentrated at one scale, stage-dependent, or divergent in total along the finite-stage tower. They affect only one positive scalar and cancel from normalized branch ratios. Uniform counting is neither required nor implied.

9.3. The admissible class is defined before cancellation

Definition 5.1 uses positivity, additivity and positive homogeneity, base locality, spatial translation covariance, fiber symmetry, and finite nontrivial output. It does not mention $|c_i|^2$, probability, product form, or projective equivalence. Product form is derived from bounded positive-operator theory, the centralizer identity, Wendel's theorem, and locality.

9.4. The role of the first-selection condition

The condition

$$\lambda_i = w_i \Lambda \quad (9.3)$$

says that the apparatus responds identically per unit derived carrier content to alternatives distinguished only by the resolved outcome label. It allows arbitrary common timing and environmental history. It does not permit an additional branch-specific multiplier.

If instead

$$\lambda_i = b_i w_i \Lambda, \quad (9.4)$$

then

$$\mathbb{P}(I = i \mid \text{click}) = \frac{b_i w_i}{\sum_j b_j w_j}, \quad (9.5)$$

not w_i in general. The branch-blind hypothesis is therefore substantive and falsifiable.

9.5. *The failed broad theorem as a self-test*

The finite-Borel counterexample demonstrates that the audit was capable of rejecting a superficially plausible theorem. The final result does not hide that failure by adding product form or continuity as an axiom. It returns to the physical L^1 carrier domain and proves the theorem there. The matrix-carrier route was likewise rejected when its attachment to the physical current could not be earned.

9.6. *Dependency chain*

The exact logical order is:

$$\begin{aligned}
 \text{Plenum quotient and bounded distinctions} &\Rightarrow \text{finite transport-invariant ledger;} \\
 L^1 \text{ carrier plus lift symmetries} &\Rightarrow \mathcal{L}(\rho) = \rho \, dx \otimes \mu; \\
 \text{product lift} &\Rightarrow [p_* \mathcal{L}(\rho)] = [\rho \, dx]; \\
 \text{local outcome projections} &\Rightarrow w_i = |c_i|^2; \\
 \text{branch-blind first selection} &\Rightarrow \mathbb{P}(I = i \mid \text{click}) = w_i.
 \end{aligned} \tag{9.6}$$

No step assumes the conclusion of a later step. The derivation is not assumption-free. Its claim is that the stated assumptions are independent of the desired probability values and jointly sufficient to produce them.

10. **Structural Significance and Relation to Prior Approaches**

The result above is framework-relative. This is an exact description of its scope, not a special defect. Quantum-foundations programs characteristically begin with a specified mathematical and physical structure and ask what follows from it.

Bohm's hidden-variable formulation supplements the wavefunction with variables and dynamics that determine individual measurement results. Everett's relative-state formulation retains universal quantum dynamics without a fundamental collapse. GRW modifies the dynamics by introducing spontaneous stochastic localization. Gleason begins with measures on closed Hilbert-space

subspaces. Envariance and decision-theoretic programs begin with entanglement symmetries or rationality conditions within their respective frameworks. None derives quantum mechanics from no premises. Each isolates a structure and determines its consequences [6, 7, 8, 9, 10, 11, 12, 13].

10.1. Gleason-type results

Gleason characterizes countably additive measures on the closed subspaces of separable real or complex Hilbert spaces of dimension at least three, showing that they have positive-operator form. Applied to quantum mechanics, the theorem supplies the density-operator form of a noncontextual measure on projection subspaces. Its input is already a measure satisfying the required additivity on the Hilbert lattice [6].

The present result does not begin by assigning a probability measure to projectors. It begins with an L^1 physical carrier, a bounded infinitesimal ledger, and a local covariant lift. The product-lift theorem determines the carrier ray before probability is introduced. The present theorem is not a replacement for Gleason's theorem and does not improve its general Hilbert-space classification.

10.2. Envariance

Zurek's envariance program seeks the Born weights from symmetries of entangled system-environment states. Environment-assisted invariance establishes equiprobability for envariantly swappable alternatives in perfectly entangled states, with further steps extending the result to general Schmidt-state amplitudes [7].

The present derivation does not use envariance, system-environment swaps, or equiprobability as a starting point. Unequal infinitesimal-scale weights are permitted throughout. The Born ratios emerge because every admissible ledger weighting multiplies the full spatial carrier by one common scalar.

10.3. Decision-theoretic derivations

Deutsch argues that the practical probabilistic consequences of quantum theory can be recovered from nonprobabilistic quantum premises together with decision-theoretic assumptions [8]. Wallace develops the program within the Everett

interpretation, deriving the probability rule from decision-theoretic assumptions about subjective likelihood [9].

Those derivations concern rational weighting or subjective likelihood within unitary quantum mechanics. The present theorem does not appeal to utilities, preferences, or rationality axioms. It derives objective carrier shares and then applies a physical competing-event process.

10.4. Bohmian hidden-variable dynamics

Bohm's 1952 formulation supplements the wavefunction with particle-position variables whose causal dynamics determine individual measurement outcomes. The actual configuration specifies the result of a measurement interaction [10, 11].

The present framework introduces no actual particle configuration or trajectory as the selector. Its primitive object in Part II is a conserved carrier density together with its Plenum fiber lift. Selection is stochastic at the first-event interface rather than deterministic through a hidden configuration.

10.5. Everett's relative-state formulation

Everett formulates quantum mechanics in terms of relative states and treats observers and measurement interactions within the universal quantum formalism without an external collapse postulate [12].

The present theorem is explicitly conditional on one first event occurring and asks which resolved alternative supplies that event. The later effective realization is intended to initiate one local record, not to interpret all branches as continuing relative records.

10.6. GRW-type objective collapse

Ghirardi, Rimini, and Weber modify the dynamics by introducing spontaneous stochastic localizations. With suitable parameters, the model leaves microscopic behavior approximately quantum mechanical while rapidly suppressing macroscopic superpositions [13].

GRW supplies an objective-collapse dynamics. The Born theorem proved here is not a stochastic modification of the Schrödinger equation. Its central result concerns the projective distribution of a Plenum-associated carrier and the conditional mark probabilities of a branch-blind first event. The absolute event

rate remains undetermined. Part III presents an illustrative local realization, but that realization is conditional and does not constitute the theorem.

10.7. *Infinitesimal structure in prior discussion*

Infinitesimal probabilities have appeared in earlier quantum-foundations work. Barrett and Goldbring construct an Everettian model with hyperfinitely many worlds in which individual branches may carry infinitesimal probabilities; they also note that Wallace’s decision-theoretic representation framework excludes positive infinitesimal probabilities [9, 14]. The present approach differs from both uses. It does not append hyperreal probabilities to an otherwise standard Hilbert-space theory. It begins from a framework in which the real line is quotient-constructed and the infinitesimal distinctions beneath each real point are structural features of the number system.

The novelty claim is correspondingly narrow: among the principal derivations surveyed here and the additional nonstandard treatment just noted, no prior Born-rule derivation has been identified that begins from quotient-constructed reals with genuine underlying infinitesimal fibers. The paper does not claim that infinitesimals have never appeared in quantum foundations.

10.8. *Structural novelty*

The novel chain is:

$$\begin{aligned}
 &\text{quotient kernel fibers} \Rightarrow \text{finite bounded relative distinctions;} \\
 &\quad \text{local covariant } L^1 \text{ lift} \Rightarrow \text{forced product form;} \\
 &\text{arbitrary fiber weighting} \Rightarrow \text{one projective carrier ray;} \\
 &\quad \text{local branch resolution} \Rightarrow w_i = |c_i|^2; \\
 &\text{branch-blind first selection} \Rightarrow \mathbb{P}(I = i \mid \text{click}) = |c_i|^2.
 \end{aligned} \tag{10.1}$$

The theorem does not claim that the Plenum is the only structure capable of satisfying these conditions. Read abstractly, it identifies a sufficient structural pattern.

10.9. *Why use non-Archimedean structure?*

The Plenum quotient, its infinitesimal kernel, and its bounded-definability structure were developed in prior work. This paper asks what follows when that existing structure is applied to the measurement carrier [1, 2, 3].

Only a small portion of the larger construction is used:

$$\mathcal{F}_{fin}/I(\mathcal{F}) \cong \mathbb{R}, \quad K = (I(\mathcal{F}))^3, \quad E_{L,B} \text{ finite.} \quad (10.2)$$

The functional-analytic lift theorem then operates entirely on ordinary $L^1(\mathbb{R}^3)$. The non-Archimedean machinery enters at one precise point: it supplies exact distinctions erased by the real quotient.

10.10. *Explanatory result and empirical scope*

The primary result explains an established law within a specified framework. It does not yet produce a new experimentally distinguishable deviation from standard quantum statistics. Potential empirical differentiation would have to arise from the later dynamical-calibration layer. No such departure is claimed in Part II.

10.11. *Checkpoint and firewall*

Part II has established the paper's principal result:

$$\boxed{\mathbb{P}(I = i \mid \text{click}) = |c_i|^2.} \quad (10.3)$$

The result belongs to the established framework-relative mode of quantum foundations, but it is not a restatement of Gleason, envariance, decision theory, Bohmian mechanics, Everett's relative-state formulation, or GRW dynamics.

Nothing in the proof of the Born branch-selection theorem depends on the conditional effective-model realization that follows.

A reader who accepts the argument through this section has received the complete principal result. Part III is an illustrative physical realization, not a premise of the theorem.

Part III. An Illustrative and Conditional Local Measurement-Event Realization

Part II proved the framework-relative, projective single-click selection law. That theorem is complete. It does not require a particular microscopic detector dynamics, an absolute event rate, or the effective construction developed below.

This part asks a separate question: can the branch-blind first-selection interface used in Theorem 8.7 be realized by a concrete local mechanism that produces one crossing, excludes competing crossings, and initiates an irreversible record?

Compression Genesis supplies one conditional effective realization. Its role is illustrative. All numerical values in this part are calibrated within the stated effective reduction. They are not universal constants or consequences of the Plenum theorem.

11. From the Projective Theorem to the Effective Edge-Flow Bridge

The earlier development of the effective model used four regional carrier properties as working assumptions: native support, fixed total content, boundary-only regional change, and nonautonomous comoving transport. They are retained here, but they no longer rest as independent assumptions. They follow from the product-lift theorem and the ordinary continuity equation for the Dirac current.

11.1. Support and comoving current

At a fixed bounded stage, Theorem 7.2 gives

$$v_f = W v_e, \quad W = \mu(D) > 0. \quad (11.1)$$

Because W is strictly positive,

$$\text{supp } v_f = \text{supp } v_e. \quad (11.2)$$

Let j_e^μ be the conserved Dirac current and define

$$J_f^\mu = W j_e^\mu. \quad (11.3)$$

Then $\nabla_\mu J_f^\mu = 0$.

Proposition 11.1. Derived Regional Edge Flow

For every sufficiently regular region U ,

$$F_U(t) = \int_U J_f^0(t, x) d^3x \quad (11.4)$$

satisfies

$$\frac{dF_U}{dt} = - \int_{\partial U} \mathbf{J}_f \cdot \mathbf{n} dS. \quad (11.5)$$

Consequently, total carrier content is conserved whenever the Dirac carrier is conserved; regional content changes only through boundary flux; the carrier is transported with the Dirac current; and no independent creation, destruction, or autonomous transport term is introduced.

11.2. Supersede and retain

The four effective properties can now be restated as derived consequences:

$$\begin{aligned} \text{native support: } & \text{supp } v_f = \text{supp } v_e; \\ \text{fixed content: } & \int J_f^0 d^3x = W \int j_e^0 d^3x; \\ \text{boundary-only change: } & \frac{d}{dt} \int_U J_f^0 d^3x = - \int_{\partial U} \mathbf{J}_f \cdot \mathbf{n} dS; \\ \text{comoving transport: } & J_f^\mu = W j_e^\mu. \end{aligned} \quad (11.6)$$

The final dependency is

Plenum product lift $\Rightarrow$ derived edge-flow properties $\Rightarrow$ regional carrier flow.

(11.7)

The properties remain useful at the effective level, but their foundation has moved downward into Part II.

11.3. Branch-resolved carrier flow

Let the alternatives be resolved by the local projections of Section 8. Their carrier contents are

$$F_i = W |c_i|^2. \quad (11.8)$$

If an interaction region recruits carrier content through its boundary, the branch-resolved incoming currents inherit the same normalized shares whenever the interaction couples identically per unit carrier and introduces no branch-dependent multiplier.

11.4. Interface boundary

The foundational theorem determines which branch share is available. It does not determine whether or when the detector crosses. Those are distinct outputs. Compression Genesis addresses the second question and is required to preserve the first answer rather than generate a new branch weighting.

12. Compression Genesis as a Conditional Local Realization

Compression Genesis is an effective local model of the onset of a measurement record. It is not presented as a fundamental replacement for quantum field theory or as a complete microscopic detector theory.

Its target is narrower:

$$\begin{aligned} \text{one weighted local crossing} &\Rightarrow \text{rapid exclusion of competitors} \\ &\Rightarrow \text{one irreversible local record seed.} \end{aligned} \quad (12.1)$$

12.1. Effective degrees of freedom

The local reduction contains a recruitment variable driven by incoming Plenum-associated carrier flow, a self-binding response that permits a recruited configuration to become locally self-holding, and a competitive response shared by the alternatives so that growth of one precursor suppresses the others.

The branch-dependent input is restricted to

$$w_i = |c_i|^2. \quad (12.2)$$

All detector and environmental coefficients are common across alternatives before selection.

12.2. Coupled-action reduction

The coupled action was constructed so that local self-binding and inter-branch competition arise from one conserved response sector rather than from independently inserted mechanisms. Elimination of the shared response gives

$$\Gamma\mu_{SN} = 0.190437 \frac{g^4 \chi_f^2}{\eta_\Phi \lambda_6}. \quad (12.3)$$

Proposition 12.1. Calibrated Coupled-Response Reduction

Within the specified effective action and normalization conventions, self-binding and competition arise from the same response, elimination of the auxiliary response gives (12.3), and the characteristic radius cancels from the reduced control parameter.

The coefficient 0.190437 is calibrated within this effective reduction. It is not a universal constant and is not part of the Born theorem.

12.3. Binary exclusivity

Let β denote the cooperative gain available to one precursor and κ the effective competitive threshold. The binary architecture has

$$\beta = 2a, \quad \kappa = a + \frac{2}{\chi_R V_R}. \quad (12.4)$$

A nontrivial precursor requires $\beta > \kappa$. Strong exclusivity requires $\kappa < \beta \leq 2\kappa$. The calibrated reduction gives

$$\frac{\beta}{\kappa} = V_R I_4 = 1.94. \quad (12.5)$$

Proposition 12.2. Calibrated Binary Exclusivity

Within the calibrated reduction, one precursor can enter the self-holding sector, while two or more simultaneous precursors are inadmissible under the retained multiplicity rule.

The value 1.94 is a calibrated model result. It is conditional on the effective action, normalization, and reduction.

12.4. One crossing and rapid lockout

Let $r_{\perp} > 0$ be the response pole of the internal competitive mode. Once one branch crosses into the precursor sector, the shared response shifts the remaining alternatives away from their crossing conditions. The calibrated worst-case estimate is

$$\boxed{t_{lock} \leq \frac{0.785596}{r_{\perp}}.} \quad (12.6)$$

The coefficient 0.785596 is calibrated within the effective response model. It is not a universal detector timescale.

Define

$$\epsilon_{ad} = \frac{\Gamma \mu_{SN}}{r_{\perp}}. \quad (12.7)$$

The operating regime requires $\epsilon_{ad} \ll 1$, so the shared competitive response acts faster than the slow precursor evolution and the window for a competing crossing remains parametrically small.

12.5. Preservation of the Born weights

Before the first crossing, the effective alternatives receive the derived shares $w_i = |c_i|^2$. The environment may modify the total timing intensity through a common history-dependent factor:

$$\lambda_i(t \mid \mathcal{H}_t) = w_i \Lambda_{CG}(t \mid \mathcal{H}_t). \quad (12.8)$$

The common factor may contain overlap, depletion, colored weak noise, shared response, pulse entry and exit, and pre-crossing memory. It may not contain an independent branch-specific coefficient.

Compression Genesis does not derive the branch weights a second time. It preserves the theorem-derived shares while supplying an exclusive physical event.

12.6. Compound transition and record initiation

After the first crossing and lockout, the precursor enters a self-holding regime. The local compound transition has three linked consequences: the selected precursor seals into a localized configuration; it separates dynamically from the parent response; and energy and phase information leak into the environmental continuum.

The endpoint is a stable local record seed whose reversal would require recovery of information already dispersed into uncontrolled environmental degrees of freedom. This is the onset of irreversibility, not a theory of observers or a complete account of detector electronics.

12.7. Logical role of the realization

The relation between Parts II and III is one-way:

$$\text{Born theorem} \Rightarrow \text{weights supplied to the effective model.} \quad (12.9)$$

There is no reverse dependency. A failure of a particular calibrated parameter, response kernel, or transition realization would require revision of this illustrative model. It would not alter Theorems 5.7, 7.2, 8.4, 8.6, or 8.7.

13. Verified Existence Witnesses

Sections 11 and 12 specify the analytic structure required of the illustrative realization. They do not by themselves show that all required inequalities can be satisfied simultaneously.

This section records explicit numerical witnesses establishing that the effective regime is nonempty. These are existence results, not universal theorems.

13.1. Status of a numerical witness

A numerical witness establishes existence by supplying one explicit parameter tuple and verifying that it satisfies a predetermined system of inequalities. The inequalities must be fixed before the tuple is accepted. Numerical search is not permitted to redefine completion, exclusivity, irreversibility, or successful event formation after the fact.

13.2. Broadband-continuum compatibility

The first potential obstruction was whether one environmental continuum could provide rapid response, weak branch-blind pre-event noise, negligible probability of a competing crossing, and irreversible leakage after the selected transition.

13.2.1. Verified Existence Result 13.1. Compatible Continuum Witness

There exists a calibrated parameter tuple Θ_{cont} for one broadband Drude-Ohmic continuum such that, within the effective reduction, the response is fast relative to precursor evolution, the adiabatic parameter is small, the pre-event noise is weak and branch-blind, the competing-crossing probability during lockout is negligible under the calibrated tolerance, and the same continuum accepts a nonzero irreversible leakage flux.

This is a verified existence witness within the selected spectral model. It does not prove that an arbitrary bath or detector has the same compatibility.

Table 13.1. Standalone continuum compatibility witness W-2.

Quantity	Verified value	Witness status
Time scaling	$\Gamma\mu_{SN} = 1$	Standalone continuum existence point
Continuum parameters	$\chi_{\perp} = 1, M_{\perp} = 10^{-4}, \eta_{\perp} = 0.03, \Omega = 100$	Drude-Ohmic class
Overdamped condition	$\eta_{\perp}^2 = 9 \times 10^{-4} > 4M_{\perp}/\chi_{\perp} = 4 \times 10^{-4}$	Satisfied
Slow response pole	$r_{\perp} = 38.197$	Fast relative to $\Gamma\mu_{SN}$
Adiabatic ratio	$\epsilon_{ad} = 0.02618$	Small
Correlation time	$\tau_c\Gamma\mu_{SN} = 0.01$	Markov-compatible at $\Omega = 100$

Quantity	Verified value	Witness status
Weak-noise ratio	$\Delta U/D = 10$	Declared weak-noise regime
Lockout exposure	$\Lambda_{\max} t_{\text{lock}} \approx 1.110 \times 10^{-7}$	Negligible competing-crossing exposure
Small-inertia headroom	2500×	Ordinary nonempty regime, not tuning

13.3. Compact-support transit

The final calibration uses the smooth compact-support pulse

$$S(t) = S_0 \cos^2 \left(\frac{\pi t}{2T} \right), \quad |t| \leq T, \quad (13.1)$$

with $S(t) = 0$ outside the transit. The selected dimensionless calibration is

$$S_0 = 0.5, \quad T = 2.5. \quad (13.2)$$

Let d denote the dimensionless upstream supply coefficient.

13.3.1. Verified Existence Result 13.2. Finite-Transit Thresholds

For the pulse above, the backward-derived minimum supply coefficient permitting any finite local measurement event is

$$\boxed{d_{\text{event}} = 2.072790004.} \quad (13.3)$$

Entry into the calibrated odd Z_2 path sector requires

$$\boxed{d > 2.078334379.} \quad (13.4)$$

At the lower finite-event threshold, the full available carrier content is

$$\boxed{F_{\text{available}} = 2.597917974.} \quad (13.5)$$

These are calibrated thresholds for the compact-support transit reduction. They are not universal physical constants.

13.4. End-to-end compact-support transit witness

Let Θ_{CST} denote the explicit compact-support transit parameter tuple reported in Appendix F.

13.4.1. Verified Existence Result 13.3. End-to-End Local Event Witness

There exists a calibrated integrated parameter tuple for which one compact transit produces carrier recruitment, one threshold crossing, competitive lockout, one self-holding precursor, one compound transition, sealing and release, and irreversible environmental leakage. All stages complete before the compact forcing pulse terminates.

The witness proves that the integrated inequalities have a nonempty intersection. It does not prove uniqueness, robustness under arbitrary perturbation, or direct correspondence with a specified laboratory detector.

Table 13.2. End-to-end compact-support witness W-3.

Block	Verified value
Scaling and selection	$\Gamma\mu_{SN} = 5.02044 \times 10^5$; $H_{sel} = 5$; $P_{select} = 1 - e^{-5} = 0.993262$
Noise	$\Delta U/D = 8$
Kernel and lockout	$r_{\perp} = 4.01288 \times 10^7$; $\epsilon_{ad} = 0.012511$; $t_{lock} \leq 1.958 \times 10^{-8}$; $P_2 < 3.92 \times 10^{-7}$
Sealed minimum	$\xi^* = 1.983657$; $z_j = 0.991795$; Hessian eigenvalues $\{1.86671, 1.96731, 1.96731, 31.29888\}$
Timing	$t_{click} = 2.2391$; $t_{settle} = 0.0951$; sealing at $2.3342 < t_{end} = 2.5$; slack 0.1658
Energy and leakage	$\Delta E_{return} = 4.152763$; $E_{int} = 3.752763$; $E_{residual} = 0.00494$ after $\gamma_E = 40$

W-2 and W-3 are independent existence witnesses. W-2 uses the standalone time unit $\Gamma\mu_{SN} = 1$, whereas W-3 uses the integrated scaling $\Gamma\mu_{SN} = 5.02044 \times 10^5$. The difference is intentional and is not a calibration inconsistency.

13.5. What the witnesses establish

Together, the witnesses establish that the conditional realization is dynamically nonempty. They do not establish that the action is uniquely selected by the Plenum, that its parameters are universal, that every detector lies in the demonstrated regime, that the regime persists under all perturbations, or that the effective variables have already been derived from a complete microscopic detector theory.

13.6. Relation to the Born theorem

The witnesses use the theorem-derived shares $w_i = |c_i|^2$ as inputs. They do not contribute to the proof that these shares equal the squared amplitudes. A failure to reproduce a witness would challenge the conditional realization, not the Born theorem.

14. Scope of the Illustrative Realization and Remaining Physical Questions

The principal theorem is complete in Part II. This section concerns only the scope of the illustrative realization and the steps required to connect it more closely to concrete detector physics.

14.1. No universal detector claim

Compression Genesis is one effective realization of the branch-blind first-selection interface. The paper does not claim that all detectors use this mechanism, that all detector materials share the same coefficients, that the compact-support calibration is universal, or that every measurement interaction admits the same reduced degrees of freedom.

The demonstrated claim is existential: at least one internally consistent effective regime realizes weighted first crossing, binary exclusion, finite-time transition, and irreversible local record initiation.

14.2. No complete microscopic derivation

The effective action is not yet derived uniquely from a complete microscopic quantum field theory of a detector and its environment. The calibrated reduction identifies a consistent dynamical architecture. A future microscopic derivation

would have to map a concrete detector model onto its effective variables and coefficients.

14.3. *Endpoint of the model*

The modeled endpoint is a stable local record seed. The paper does not attempt to model complete detector amplification, electronic readout, external storage, observer awareness, or the full thermodynamic history of a macroscopic record.

14.4. *Branch blindness as a physical condition*

Theorem 8.7 requires

$$\lambda_i = w_i \Lambda. \quad (14.1)$$

If an apparatus introduces an additional outcome-dependent response b_i , then

$$\lambda_i = b_i w_i \Lambda, \quad (14.2)$$

and the theorem does not predict unmodified Born selection for those effective labels. A physical application must identify the correctly resolved outcome sectors and verify that no unaccounted branch-dependent coupling remains.

14.5. *Robustness beyond a single witness*

The explicit compact-support witness establishes nonemptiness. A stronger physical case requires a neighborhood of successful parameter values, sensitivity analysis for the continuum and response parameters, quantitative control of competing-event probability under colored noise, stability under pulse perturbations, an ensemble of successful trajectories, and a concrete detector mapping.

These are hardening tasks for the illustrative realization. They are not missing steps in the Born theorem.

14.6. *Experimental contact*

The central result reproduces the established Born law. It does not presently predict a deviation from standard quantum statistics. Potential empirical contact would have to arise from the dynamical realization, such as finite lockout times,

detector-dependent leakage profiles, departures from ideal branch blindness, rare competing events near the exclusivity boundary, or pulse-duration thresholds. No such signature is claimed as a prediction in this paper.

14.7. *External validation*

The proofs and calculations have been subjected to internal adversarial checking, including counterexample searches and reconstruction of the load-bearing functional-analytic argument. They have not yet received independent specialist review by the quantum-foundations, measure-theory, or non-Archimedean mathematics communities.

The appropriate status is:

internally formalized and audit-passed; external review pending.	(14.3)
--	--------

A defect in Part III would not by itself invalidate Part II. A defect in the product-lift, projective-cancellation, or first-selection proofs would bear directly on the central theorem.

15. **Conclusion**

This paper began with a question: whether the Born pattern must remain primitive. Quantum theory states, with a precision matched almost nowhere else in science, how often each alternative is realized. But precision is not explanation, and for a century the source of that pattern was left unspoken, as though a rule confirmed often enough no longer needs an origin. The result established here is that, within the Plenum framework, it has one.

The principal derivations surveyed in this paper work within the ordinary real continuum, where a real number is a featureless point with nothing underneath it, and where the squared-amplitude law, having no deeper structure to rest on, could only be characterized or postulated. The derivation given here was possible because it was carried out in a framework of a different kind: one in which the real line is not taken as primitive but built as a quotient, so that beneath each real point there genuinely exists a structured layer of infinitesimal distinctions. To my knowledge, no previous derivation of the Born rule has been identified from

that specific vantage. It is in the structure of those erased distinctions, not in any added postulate, that the squared-amplitude law turns out to be written.

What survives the passage back to real space is only a ratio. The microscopic weights of the fiber may vary as they please; the absolute scale of the carrier is never fixed and is never needed. What is fixed is the relation between the alternatives, and it is exactly the relation the Born rule prescribes. The probability is not assumed at the outset and recovered at the end. It is determined first as a carrier share, a statement of how much of a conserved quantity each alternative holds, and only afterward inherited by the physical act of selection. That order, structure first and probability second, is what keeps the derivation free of circularity, and it is the reason the squared amplitudes can appear as a conclusion rather than as an assumption in disguise.

It would be incomplete to present that mathematics without the physical realization that made its measurement meaning clear. A carrier share is not yet a probability. The product-lift theorem fixes how much of the conserved carrier each alternative holds, and the common-history selection theorem proves that a branch-blind first event inherits those shares as its conditional probabilities. Part III then supplies a concrete local picture of how such an event can occur: one alternative crosses, its competitors are excluded, and an irreversible record begins. Those mechanics are not a premise of the mathematical proof. They are what made clear how the theorem could describe measurement rather than remain only a statement about conserved density.

I have been exact about the boundaries throughout, because the boundaries are the claim and not a retreat from it. The result holds within a particular framework, and it fixes the ratios of outcomes rather than any absolute scale. But framework-relativity is the ordinary condition of foundational work, and a rule about ratios is captured completely by a derivation about ratios. The physical realization offered in Part III is more tentative still, one instance among the many the theorem permits, and nothing in the theorem depends upon it. What the theorem itself asserts is narrow and, I believe, secure: under the explicit hypotheses of Theorem 8.7, the probability of an outcome is the square of its amplitude.

There is further work, and it is honest to name it. The effective variables must be connected to real detector systems; the single successful trajectory must be widened into a robust family; and whether any of this leaves a trace that an experiment could find remains to be seen. None of that is settled here, and none

of it is concealed. But the question with which the paper began is answered within the structure it uses. The pattern that governs quantum outcomes need not be placed beyond explanation simply because it was measured for a century without its source being found. Within the structure developed here it is no longer primitive. It is the projective consequence of distinctions that ordinary real space has erased.

Author's Note

This is my eighth paper. It addresses a problem at the foundations of physics: the origin of the squared-amplitude weights that Born's rule assigns to quantum outcomes. The derivation draws on mathematical work I developed earlier, in which the real line arises as a quotient and the distinctions ordinarily erased beneath each real point are retained as genuine objects. That framework was built as mathematics, with no application to quantum theory in view, and this paper is where it unexpectedly proved useful. The result reported here is what came of that encounter.

The work was developed and tested with several computational systems used in distinct and deliberately adversarial roles: to formalize arguments, perform calculations, stress-test proofs, and above all attempt to break one another's results. Every retained load-bearing proof and calculation was subjected to independent scrutiny, and approaches that failed were discarded rather than repaired into the appearance of success. I directed that process, reviewed and edited every part of this paper, and remain solely responsible for its content and for any error that remains. The tools were instruments; the judgments, and the answerability for them, are mine.

It is a century since Max Born introduced the rule that assigns quantum probabilities their weights, and nearly as long since that rule prompted Einstein's objection that God does not play dice. This paper does not settle whether quantum chance is fundamental. It does, however, offer a way of understanding where those weights come from: within the framework used here, what appears as irreducible probability is the projective consequence of structure that ordinary real space has erased. I place the paper into circulation in the hope that it will receive the scrutiny it has not yet had, because a claim of this kind should be accepted, revised, or rejected only after the strongest examination others can give it.

Appendix A. Minimal Plenum Definitions and Notation

A.1. Purpose and citation boundary

This appendix collects the notation and inherited mathematical structures used in the proof. It does not reproduce the construction of the Plenum or the bounded-definability results established in the prior papers [1, 2, 3].

Inherited from prior work are $\mathcal{F}$, $\mathcal{F}_{fin}$, $I(\mathcal{F})$, st , π , K , $E_{L,B}$, and the finite-stage Abyss construction. Introduced or proved here are $\mathcal{K}_x$, $D_{L,B}$ as a relative ledger, $\mathcal{L}$, μ , the product-lift theorem, projective cancellation, and the single-click selection theorem.

A.2. The Plenum and its finite part

Let $\mathcal{F}$ denote the Plenum Field constructed in prior work by an iterated ultrapower and direct-limit procedure [1, 2]. The finite part is

$$\mathcal{F}_{fin} = \{x \in \mathcal{F} : |x| < n \text{ for some positive standard integer } n\}. \quad (\text{A.1})$$

Its infinitesimal ideal is

$$I(\mathcal{F}) = \left\{ x \in \mathcal{F} : |x| < \frac{1}{n} \text{ for every positive standard integer } n \right\}. \quad (\text{A.2})$$

The standard-part map is a surjective ordered-ring homomorphism

$$st : \mathcal{F}_{fin} \rightarrow \mathbb{R}, \quad (\text{A.3})$$

with kernel $I(\mathcal{F})$. Consequently,

$$\mathbb{R} \cong \mathcal{F}_{fin}/I(\mathcal{F}). \quad (\text{A.4})$$

No canonical representative of a real equivalence class is assumed.

A.3. Spatial quotient and fibers

Define $\mathcal{S} = \mathcal{F}^3$ and $\mathcal{S}_{fin} = \mathcal{F}_{fin}^3$. The componentwise standard-part map is

$$\pi = st_3 : \mathcal{S}_{fin} \rightarrow \mathbb{R}^3, \quad (\text{A.5})$$

with kernel

$$K = (I(\mathcal{F}))^3. \quad (\text{A.6})$$

For $x \in \mathbb{R}^3$, the fiber beneath x is

$$\mathcal{K}_x = \pi^{-1}(x) = X + K \quad (\text{A.7})$$

for any X with $\pi(X) = x$. It is an affine coset with no preferred origin.

A.4. Bounded definability

Let L be an admissible finite-symbol language and B a finite symbol budget. Define $E_{L,B}$ as the set of positive infinitesimals uniquely definable by an L -formula of length at most B . Prior work establishes that $E_{L,B}$ is finite [3]. Where nonempty, define

$$\Gamma_{L,B} = \min E_{L,B}. \quad (\text{A.8})$$

If $\Lambda_{L,B}$ is the maximum uniquely definable positive infinite element in the same bounded presentation, the corresponding Abyss element is

$$A_{L,B} = \frac{\Gamma_{L,B}}{\Lambda_{L,B}}, \quad (\text{A.9})$$

with $0 < A_{L,B} < \epsilon$ for every $\epsilon \in E_{L,B}$. The construction is presentation-relative. No absolute smallest positive infinitesimal is asserted.

A.5. Relative-distinction ledger

For each $\epsilon \in E_{L,B}$,

$$C_\epsilon = \{(\sigma_1\epsilon, \sigma_2\epsilon, \sigma_3\epsilon) : \sigma_i \in \{-1, +1\}\}. \quad (\text{A.10})$$

Because $\epsilon > 0$, the eight sign choices produce eight distinct elements. The bounded relative-distinction ledger is

$$D_{L,B} = \bigcup_{\epsilon \in E_{L,B}} C_\epsilon. \quad (\text{A.11})$$

The signed-coordinate and coordinate-permutation group is

$$G_D = \{-1, +1\}^3 \rtimes S_3. \quad (\text{A.12})$$

A general finite positive G_D -invariant measure on $D_{L,B}$ has the form

$$\mu = \sum_{\epsilon \in E_{L,B}} a_\epsilon \#_{C_\epsilon}, \quad a_\epsilon \geq 0. \quad (\text{A.13})$$

Uniform cross-scale weighting is not assumed.

A.6. Carrier and measure notation

The ordinary spatial base is $X = \mathbb{R}^3$ with Lebesgue measure $dx = d^3x$. For $\rho \in L_+^1(X)$, define

$$v_\rho(A) = \int_A \rho(x) dx. \quad (\text{A.14})$$

An admissible ledger lift is

$$\mathcal{L} : L_+^1(X) \rightarrow \mathcal{M}_+(X \times D). \quad (\text{A.15})$$

The bundle projection is $p(x, d) = x$. For nonzero finite positive measures ν and ν' , projective equivalence is

$$\nu \sim \nu' \Leftrightarrow \nu' = a\nu \text{ for some } a > 0. \quad (\text{A.16})$$

The corresponding ray is $[\nu] = \{a\nu : a > 0\}$.

A.7. Dirac carrier notation

Let

$$\Psi \in L^2(X, \mathbb{C}^4 \otimes \mathcal{V}) \quad (\text{A.17})$$

be a normalizable Dirac carrier state, where $\mathcal{V}$ is a finite-dimensional internal multiplicity space when internal alternatives are present. Its carrier density is

$$\rho_e = j_e^0 = \Psi^\dagger \Psi. \quad (\text{A.18})$$

The current satisfies $\nabla_\mu j_e^\mu = 0$. At a fixed bounded stage,

$$J_f^\mu = W j_e^\mu, \quad W = \mu(D) > 0. \quad (\text{A.19})$$

The subscript f is retained only as a compact label for the Plenum-associated carrier. It does not denote an independent autonomous field.

A.8. Outcome projections and branches

Let $P_1(x), \dots, P_m(x)$ be measurable orthogonal projection fields with $P_i P_j = \delta_{ij} P_i$ and $\sum_i P_i = I$ on the occupied outcome subspace. For

$$\Psi = \sum_i c_i \Phi_i, \quad (\text{A.20})$$

assume $\|\Phi_i\|_2 = 1$, $\sum_i |c_i|^2 = 1$, and $P_i \Phi_j = \delta_{ij} \Phi_j$. The projected carrier density is

$$\rho_{e,i} = \Psi^\dagger P_i \Psi. \quad (\text{A.21})$$

Its Plenum-associated content is $F_i = p_* \mathcal{L}(\rho_{e,i})(X)$ and the normalized carrier share is

$$w_i = \frac{F_i}{\sum_j F_j} = |c_i|^2. \quad (\text{A.22})$$

Equation (A.19) is first a carrier-share identity. Its probabilistic interpretation enters only through the first-selection process.

A.9. First-selection notation

Let T be the time of the first local measurement event and $I \in \{1, \dots, m\}$ its outcome mark. The branch-blind common-history condition is

$$\lambda_i(t \mid \mathcal{H}_t) = w_i \Lambda(t \mid \mathcal{H}_t). \quad (\text{A.23})$$

Under Theorem 8.6,

$$\mathbb{P}(I = i \mid T < \infty) = w_i. \quad (\text{A.24})$$

Combining (A.19) and (A.21) gives the single-click law.

A.10. Conventions

Unless stated otherwise, equalities between L^1 functions hold almost everywhere; all measures are finite positive Borel measures; bounded-ledger statements are made at a fixed finite stage; no completed infinite-ledger measure is assumed; no absolute Plenum carrier normalization is fixed; “projective” refers to equality of positive measure rays; spatial separation means disjoint support up to null sets; internal-mode separation means resolution by mutually orthogonal local projections; a single click means the first marked local measurement event conditioned on an event occurring; and frequency statements require assumptions additional to the single-event theorem.

Appendix B. Complete Proof of the L^1 Product-Lift Theorem

B.1. Statement and setup

Let $X = \mathbb{R}^3$ with Lebesgue measure and let D be a finite bounded relative-distinction ledger. Consider

$$\mathcal{L} : L_+^1(X) \rightarrow \mathcal{M}_+(X \times D). \quad (\text{B.1})$$

Assume additivity and positive homogeneity, base locality, spatial translation covariance, fiber-symmetry invariance, and finite nontrivial output, exactly as in Definition 5.1.

Theorem B.1. L^1 Product-Lift Theorem

There exists a unique finite positive measure μ on D , independent of ρ , such that

$$\boxed{\mathcal{L}(\rho) = \rho(x) dx \otimes \mu} \quad (\text{B.2})$$

for every $\rho \in L^1_+(X)$. The measure is G_D -invariant and satisfies $0 < \mu(D) < \infty$. Product form, operator boundedness, continuity, module-linearity, and projective cancellation are not hypotheses.

B.2. Slice measures

For each $d \in D$ and $\rho \in L^1_+(X)$, define

$$\mathcal{L}_d(\rho)(A) = \mathcal{L}(\rho)(A \times \{d\}). \quad (\text{B.3})$$

Lemma B.2. Absolute Continuity of the Slices

For every d and ρ ,

$$\mathcal{L}_d(\rho) \ll \nu_\rho \ll dx. \quad (\text{B.4})$$

Proof. If $\nu_\rho(A) = 0$, then $\rho = 0$ almost everywhere on A . Base locality gives $\mathcal{L}(\rho)(A \times D) = 0$, and positivity gives $\mathcal{L}_d(\rho)(A) = 0$. $\square$

By Radon-Nikodym, there is a unique $T_d\rho \in L^1_+(X)$, up to equality almost everywhere, such that

$$\mathcal{L}_d(\rho)(A) = \int_A (T_d\rho)(x) dx. \quad (\text{B.5})$$

B.3. Cone linearity and extension

Lemma B.3. Cone Linearity

For $a, b \geq 0$ and $\rho_1, \rho_2 \in L_+^1(X)$,

$$T_d(a\rho_1 + b\rho_2) = aT_d\rho_1 + bT_d\rho_2. \quad (\text{B.6})$$

This follows from equality of the corresponding slice measures and Radon-Nikodym uniqueness.

Lemma B.4. Positive Linear Extension

The map T_d extends uniquely to an everywhere-defined positive linear operator $T_d : L^1(X) \rightarrow L^1(X)$.

Proof. For $f = u - v$ with $u, v \geq 0$, define $T_d f = T_d u - T_d v$. If $u - v = u' - v'$, then $u + v' = u' + v$, and cone additivity proves independence of decomposition. Linearity, positivity, and uniqueness follow. $\square$

*B.4. Locality of the extended operator***Lemma B.5. Operator Locality**

If $f = 0$ almost everywhere on a Borel set A , then $T_d f = 0$ almost everywhere on A .

Proof. Both f_+ and f_- vanish on A . Base locality applied separately to their slice measures gives the result. $\square$

Equivalently,

$$\text{ess supp}(T_d f) \subseteq \text{ess supp}(f). \quad (\text{B.7})$$

B.5. Automatic boundedness

Lemma B.6. Lattice Inequality

For every $f \in L^1(X)$,

$$|T_d f| \leq T_d |f| \quad (\text{B.8})$$

almost everywhere.

Lemma B.7. Automatic Boundedness

The operator $T_d : L^1(X) \rightarrow L^1(X)$ is bounded.

Proof. Suppose T_d is unbounded. Choose $g_n \in L^1_+(X)$ with $\|g_n\|_1 = 1$ and $\|T_d g_n\|_1 \geq 2^n$. Set $\rho_n = 2^{-n} g_n$. Then $\sum_n \rho_n$ converges in $L^1_+(X)$ to some ρ , while $\|T_d \rho_n\|_1 \geq 1$.

For every N , positivity gives

$$T_d \rho \geq \sum_{n=1}^N T_d \rho_n. \quad (\text{B.9})$$

All terms are nonnegative, so

$$\|T_d \rho\|_1 \geq \sum_{n=1}^N \|T_d \rho_n\|_1 \geq N. \quad (\text{B.10})$$

This contradicts $T_d \rho \in L^1(X)$. □

The proof uses only one summed input, finite linearity, positivity, and finite output. It does not use countable additivity of $\mathcal{L}$ as an operator-valued map.

*B.6. Translation covariance and the centralizer identity***Lemma B.8. Slice Translation Covariance**

For every $v \in X$ and $f \in L^1(X)$,

$$T_d(\tau_v f) = \tau_v(T_d f). \quad (\text{B.11})$$

The identity follows by applying full-lift covariance to the slice measure and using Radon-Nikodym uniqueness.

Lemma B.9. Centralizer Identity

For all $f, h \in L^1(X)$,

$$T_d(f * h) = f * T_d h. \quad (\text{B.12})$$

Proof. Write

$$f * h = \int_X f(y) \tau_y h \, dy \quad (\text{B.13})$$

as an L^1 -valued Bochner integral. Boundedness allows T_d to pass through the integral, and covariance gives

$$T_d(f * h) = \int_X f(y) \tau_y (T_d h) \, dy = f * T_d h. \quad (\text{B.14})$$

□

B.7. Wendel representation

The additive group $G = (\mathbb{R}^3, +)$ is locally compact with Lebesgue Haar measure. By Wendel's theorem, a bounded left centralizer of $L^1(G)$ is convolution on the right by a unique regular measure of bounded variation. Therefore there is a unique finite regular measure of bounded variation η_d such that

$$T_d f = f * \eta_d, \quad (\text{B.15})$$

and $\|T_d\| = |\eta_d|(X)$.

The representation theorem used here is Wendel's centralizer theorem [4, Theorem 1].

B.8. Positivity of the representing measure

Choose a nonnegative approximate identity (e_α) with integral one. Since T_d is positive, $e_\alpha * \eta_d \geq 0$.

Lemma B.10. Positivity of η_d

The Wendel representing measure η_d is positive.

Proof. Let $\varphi \in C_c(X)$ be nonnegative. Then

$$\int_X \varphi(x) (e_\alpha * \eta_d)(x) dx \geq 0. \quad (\text{B.16})$$

Fubini gives

$$\int_X \left[\int_X \varphi(y+z) e_\alpha(z) dz \right] d\eta_d(y) \geq 0. \quad (\text{B.17})$$

The bracketed functions converge pointwise to φ and are uniformly bounded. Dominated convergence with respect to $|\eta_d|$ gives $\int \varphi d\eta_d \geq 0$. Hence η_d is positive. $\square$

B.9. Locality collapses the support**Lemma B.11. Locality-to-Point-Mass Lemma**

If $T_d f = f * \eta_d$ is local, then $\text{supp} \eta_d \subseteq \{0\}$.

Proof. Suppose $y_0 \neq 0$ belongs to the support. Choose $0 < r < 2|y_0|/5$, set $A = B(y_0, r/2)$ and $f = \mathbf{1}_{B(0, 2r)}$. The sets A and $B(0, 2r)$ are disjoint, so $f = 0$ on A . But for $x, y \in A$, $x - y \in B(0, 2r)$, and therefore

$$T_d f(x) \geq \eta_d(A) > 0 \quad (\text{B.18})$$

for every $x \in A$. This contradicts locality. $\square$

Corollary B.12. Slice Multiplication Form

There is a constant $c_d \geq 0$ such that

$$\eta_d = c_d \delta_0, \quad T_d f = c_d f. \quad (\text{B.19})$$

B.10. Reconstruction of the full lift

Define

$$\mu = \sum_{d \in D} c_d \delta_d. \quad (\text{B.20})$$

Lemma B.13. Product Reconstruction

For every $\rho \in L_+^1(X)$,

$$\mathcal{L}(\rho) = \nu_\rho \otimes \mu. \quad (\text{B.21})$$

Proof. For a Borel set $E \subseteq X \times D$, let $E_d = \{x : (x, d) \in E\}$. Since D is finite,

$$E = \bigsqcup_{d \in D} (E_d \times \{d\}). \quad (\text{B.22})$$

Summing the slice identities gives equality with the product measure. $\square$

B.11. Fiber symmetry, nontriviality, and uniqueness

Lemma B.14. Orbit Constancy

For every $g \in G_D$ and $d \in D$, $c_{g(d)} = c_d$.

Lemma B.15. Positive Total Fiber Mass

The finite nontrivial output condition implies

$$0 < \mu(D) < \infty. \quad (\text{B.23})$$

Lemma B.16. Uniqueness of the Fiber Measure

The measure μ in (B.2) is unique.

Lemmas B.2 through B.16 prove Theorem B.1.

B.12. *Dependency audit and boundary*

The proof uses positivity and locality to obtain absolutely continuous slices; cone linearity to obtain linear operators; positivity and everywhere-defined finiteness to force boundedness; translation covariance to obtain the centralizer identity; Wendel's theorem to obtain convolution; positivity to obtain a positive representing measure; locality to concentrate that measure at the origin; finiteness of D to reconstruct the product; and fiber symmetry to equalize weights within each orbit.

No step uses $|c_i|^2$, an outcome probability, equal cross-scale counting, or projective cancellation.

The theorem materially depends on the L_+^1 domain. It does not establish the same result for all finite Borel measures.

Appendix C. Counterexample on the Domain of All Finite Borel Measures

C.1. *Purpose*

A superficially broader product-lift theorem on all finite positive Borel measures is false. This appendix gives an explicit counterexample satisfying the broad analogues of the admissibility conditions while failing both universal product form and projective preservation.

C.2. *Atomic and nonatomic decomposition*

For $\nu \in \mathcal{M}_+(X)$, let

$$\nu_{at} = \sum_{x \in X} \nu(\{x\}) \delta_x, \quad \nu_{na} = \nu - \nu_{at}. \quad (\text{C.1})$$

The decomposition is unique, additive, positively homogeneous, and translation-covariant.

C.3. *Construction*

Choose finite positive G_D -invariant measures μ_0 and μ_1 on D , each with positive total mass. Define

$$\boxed{\mathcal{L}_B(v) = v \otimes \mu_0 + v_{na} \otimes \mu_1.} \quad (\text{C.2})$$

Proposition C.1. Broad Admissibility

The map $\mathcal{L}_B$ is positive, additive, positively homogeneous, base-local, spatially translation-covariant, fiber-symmetric, finite, and nontrivial.

Proof. The atomic and nonatomic maps are cone-linear and commute with translations. Base locality follows from $0 \leq v_{na} \leq v$. The total output is

$$\mu_0(D)v(X) + \mu_1(D)v_{na}(X), \quad (\text{C.3})$$

which is finite and positive for nonzero v . $\square$

C.4. Failure of universal product form

Theorem C.2. Broad-Domain Product Failure

No fixed finite measure μ on D satisfies $\mathcal{L}_B(v) = v \otimes \mu$ for every $v \in \mathcal{M}_+(X)$.

Proof. A Dirac mass forces $\mu = \mu_0$, while a nonzero nonatomic measure forces $\mu = \mu_0 + \mu_1$. Since $\mu_1 \neq 0$, these are incompatible. $\square$

C.5. Failure of projective preservation

The pushforward is

$$p_*\mathcal{L}_B(v) = \mu_0(D)v + \mu_1(D)v_{na}. \quad (\text{C.4})$$

For $v = \delta_x + \lambda$, with λ nonzero and nonatomic, the atomic part is scaled by $\mu_0(D)$ and the nonatomic part by $\mu_0(D) + \mu_1(D)$. Hence the pushforward is not proportional to v .

C.6. Collapse on the L^1 carrier domain

If $\nu_\rho = \rho dx$ with $\rho \in L^1_+(X)$, then ν_ρ is nonatomic, so

$$\mathcal{L}_B(\nu_\rho) = \nu_\rho \otimes (\mu_0 + \mu_1). \quad (\text{C.5})$$

This particular obstruction disappears on the physical carrier domain. It does not itself prove the positive theorem; Appendix B proves that every admissible L^1 lift has product form.

C.7. Logical significance

The correction is to replace an overbroad carrier domain with the actual Dirac carrier domain, not to add a condition whose purpose is to force the desired conclusion. The final theorem is narrower because the false generalization was rejected, not because its failure was hidden.

Appendix D. Common-History First Selection and Repeated-Trial Frequencies

D.1. Purpose

The projective carrier theorem fixes $w_i = |c_i|^2$ before probability is introduced. This appendix proves the separate stochastic statement converting those shares into conditional first-event probabilities.

D.2. First-event system

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$ be a filtered probability space satisfying the usual conditions. Let $T \in [0, \infty]$ be the first-event time and $I \in \{1, \dots, m\}$ its mark on $\{T < \infty\}$.

Define

$$N_i(t) = \mathbf{1}_{\{T \leq t, I=i\}}, \quad N(t) = \sum_i N_i(t) = \mathbf{1}_{\{T \leq t\}}, \quad (\text{D.1})$$

and the predictable at-risk process

$$Y(t) = \mathbf{1}_{\{T \geq t\}} = 1 - N(t-). \quad (\text{D.2})$$

The process is simple, so simultaneous first events of distinct marks have probability zero.

D.3. Cause-specific intensities

Let $\lambda_i(t \mid \mathcal{F}_{t-})$ be the cause-specific pre-event intensity. The compensated process

$$M_i(t) = N_i(t) - \int_0^t Y(s) \lambda_i(s \mid \mathcal{F}_{s-}) ds \quad (\text{D.3})$$

is a local martingale under the usual integrability conditions.

Definition D.1. Branch-Blind Common-History Factorization

The interface is branch-blind with shares w_i if $w_i \geq 0$, $\sum_i w_i = 1$, and

$$\lambda_i(t \mid \mathcal{F}_{t-}) = w_i \Lambda(t \mid \mathcal{F}_{t-}) \quad (\text{D.4})$$

for one nonnegative predictable common process Λ . The common factor may depend on shared overlap, depletion, memory, colored evolution, and pulse history. It may not contain an outcome-specific multiplier.

D.4. Compensation identity

For every nonnegative predictable process H ,

$$\mathbb{E} \left[\int_0^\infty H(t) dN_i(t) \right] = \mathbb{E} \left[\int_0^\infty H(t) Y(t) \lambda_i(t \mid \mathcal{F}_{t-}) dt \right]. \quad (\text{D.5})$$

This is the standard compensation identity for counting processes with predictable compensators; see Andersen et al. [5, Ch. II].

Taking $H = 1$ gives

$$\mathbb{P}(I = i, T < \infty) = \mathbb{E} \left[\int_0^\infty Y(t) \lambda_i(t \mid \mathcal{F}_{t-}) dt \right]. \quad (\text{D.6})$$

D.5. Common-history first-selection theorem

Theorem D.2. Common-History First-Selection Theorem

Assume the simple first-event system, the factorization (D.4), integrable compensators, and $P(T < \infty) > 0$. Then

$$\boxed{P(I = i \mid T < \infty) = w_i.} \quad (\text{D.7})$$

Proof. Substitution of (D.4) into (D.6) gives

$$P(I = i, T < \infty) = w_i \mathbb{E} \left[\int_0^\infty Y(t) \Lambda(t \mid \mathcal{F}_{t-}) dt \right]. \quad (\text{D.8})$$

Summing over i identifies the expectation as $P(T < \infty)$. Division proves the result. $\square$

The theorem assumes no independent branch clocks, no Markov property, and no deterministic or Poisson hazard.

*D.6. Time-mark factorization***Corollary D.3. Joint Time-Mark Factorization**

For every Borel set $A \subseteq [0, \infty)$,

$$P(I = i, T \in A) = w_i P(T \in A). \quad (\text{D.9})$$

Thus, conditioned on a first event occurring, the mark and event time are independent. Common history may reshape the complete time distribution without altering conditional outcome weights.

D.7. Branch-sensitive boundary

If instead

$$\lambda_i = b_i w_i \Lambda, \quad b_i > 0, \quad (\text{D.10})$$

then

$$\mathbb{P}(I = i \mid T < \infty) = \frac{b_i w_i}{\sum_j b_j w_j}. \quad (\text{D.11})$$

This counterfactual identifies the exact physical content of branch blindness.

D.8. Application to the carrier shares

Theorem 8.4 establishes $w_i = |c_i|^2$ before probability enters. Combining it with Theorem D.2 gives

$$\boxed{\mathbb{P}(I = i \mid T < \infty) = |c_i|^2.} \quad (\text{D.12})$$

The carrier theorem and probability theorem have different logical roles.

D.9. Repeated-trial frequency theorem

Let (T_n, I_n) be independent and identically distributed trials satisfying Theorem D.2 with the same shares. Define $Y_n = \mathbf{1}_{\{T_n < \infty\}}$ and $X_{n,i} = \mathbf{1}_{\{T_n < \infty, I_n = i\}}$, with $p_c = \mathbb{P}(T_n < \infty) > 0$.

Theorem D.4. Conditional Frequency Law

Almost surely,

$$\boxed{\frac{\sum_{n=1}^N X_{n,i}}{\sum_{n=1}^N Y_n} \rightarrow w_i.} \quad (\text{D.13})$$

Proof. The strong law gives $N^{-1} \sum Y_n \rightarrow p_c$ and $N^{-1} \sum X_{n,i} \rightarrow p_c w_i$. Division gives the result. $\square$

This frequency theorem assumes repeated trials, independence, identical preparation and response, common shares, and positive event probability. None is used to prove the single-event law.

Appendix E. Coupled-Response Reduction, Exclusivity, and Lockout

E.1. Purpose and status

This appendix records the analytic reductions used by the conditional local measurement-event model in Part III. The results have three distinct statuses:

1. exact algebra within the retained effective reduction;
2. calibrated identities within that reduction;
3. conditional asymptotic consequences under weak continuous branch-blind noise and fast response.

None is part of the proof of the projective Born selection theorem. The effective reduction has not been derived uniquely from a complete microscopic detector field theory. Its symbols denote reduced coefficients, not fundamental constants.

E.2. Reduced variables

The retained reduction uses positive coefficients g , χ_f , η_Φ , and λ_6 . The product $\Gamma\mu_{SN}$ is dimensionless in the adopted normalization and enters the adiabatic consistency condition. The binary competition variables are β , κ , V_R , and I_4 . The internal response pole is $r_\perp > 0$.

E.3. Shared-response reduction

The model uses one response sector for precursor self-binding and intermode competition. Eliminating the shared response and evaluating the retained radial integrals gives

$$\Gamma\mu_{SN} = 0.190437 \frac{g^4 \chi_f^2}{\eta_\Phi \lambda_6}. \quad (\text{E.1})$$

Proposition E.1. Coupled-Response Reduction

Within the retained reduction, the same eliminated response supplies self-binding and intermode competition, the precursor-radius dependence cancels from the product $\Gamma\mu_{SN}$, and the reduced product is given by (E.1).

The radius cancellation is exact within the retained reduction. The coefficient 0.190437 is the calibrated evaluation of the selected radial integrals. It is not a universal constant and is not asserted to persist under a different radial ansatz, action, or detector model.

E.4. Binary-mode reduction

Let $a > 0$ and define

$$\beta = 2a, \quad \kappa = a + \frac{2}{\chi_R V_R}. \quad (\text{E.2})$$

The one-precursor existence condition is $\beta > \kappa$.

Lemma E.2. Existence Implies the Strong Binary Window

Under (E.2), $\beta > \kappa$ implies

$$\kappa < \beta < 2\kappa. \quad (\text{E.3})$$

Proof. The lower inequality is the hypothesis. Since $2\kappa = 2a + 4/(\chi_R V_R) > 2a = \beta$, the upper inequality follows. $\square$

E.5. Multiplicity criterion

The retained m -precursor admissibility condition is

$$\beta > m\kappa. \quad (\text{E.4})$$

Theorem E.3. Binary Exclusivity

Within this multiplicity rule, one-precursor existence implies that every $m \geq 2$ state is inadmissible.

Proof. Lemma E.2 gives $\beta < 2\kappa \leq m\kappa$ for every $m \geq 2$. $\square$

This is exact algebra within the binary reduction. It does not use the calibrated value of β/κ .

E.6. Calibrated exclusivity ratio

The calibrated geometry gives

$$\frac{\beta}{\kappa} = V_R I_4 = 1.9400860506, \quad (\text{E.5})$$

reported in the body as 1.94. Since $1 < 1.9400860506 < 2$, the calibration lies strictly inside the strong exclusivity window.

The number is model-relative. A different radial profile or response reduction may produce a different ratio or may fail to enter the interval.

E.7. Finite response and lockout

Write the normalized competitive response after the first crossing in the reduced relaxation form

$$z(t) = z_\infty \left[1 - e^{-r_\perp(t-t_*)} \right], \quad t \geq t_*. \quad (\text{E.6})$$

If $\theta_{lock} = z_{lock}/z_\infty$, then

$$t_{lock} = -\frac{1}{r_\perp} \ln(1 - \theta_{lock}). \quad (\text{E.7})$$

The calibrated worst-case threshold gives

$$\boxed{t_{lock} \leq \frac{0.785596}{r_\perp}}. \quad (\text{E.8})$$

The coefficient 0.785596 is calibrated within the response model and is not a universal detector timescale.

E.8. Double-crossing bound

If branch i crosses first, the probability that another branch crosses during lockout is bounded by the integrated competing intensity:

$$P_{2|1} \leq \int_{t_*}^{t_*+t_{lock}} \sum_{j \neq i} \lambda_j(t \mid \mathcal{H}_t) dt. \quad (\text{E.9})$$

Under branch blindness, $\lambda_j = w_j \Lambda$, and if $\Lambda \leq \Lambda_{\max}$ throughout lockout,

$$P_{2|1} \leq 0.785596 (1 - w_i) \frac{\Lambda_{\max}}{r_{\perp}}. \quad (\text{E.10})$$

This is a bound, not an assertion of identically zero probability.

E.9. Adiabatic consistency

Define

$$\epsilon_{ad} = \frac{\Gamma \mu_{SN}}{r_{\perp}} = \frac{0.190437}{r_{\perp}} \frac{g^4 \chi_f^2}{\eta_{\Phi} \lambda_6}. \quad (\text{E.11})$$

The regime $\epsilon_{ad} \ll 1$ states that the internal competitive response is fast relative to the nonlinear precursor scale. Under weak continuous branch-blind noise and no first-order simultaneous multi-branch shock, the probability of a competing crossing during lockout is parametrically negligible.

E.10. Branch-blindness boundary

Permitted pre-event factors include shared response renormalization, common barrier evolution, common depletion, common colored noise, and common pulse dependence. An additional branch-specific factor b_i changes the normalized outcome law and falls outside the theorem's hypotheses. A scalar or energetic term that modifies the precursor barrier is admissible only if its effect is common across equivalent alternatives.

E.11. Status summary

Result	Status	Scope
Shared response supplies binding and competition	Exact within retained reduction	Not a microscopic uniqueness claim
Radius cancellation in $\Gamma \mu_{SN}$	Exact within retained reduction	Depends on retained radial structure
Coefficient 0.190437	Calibrated	Not universal

Result	Status	Scope
$\kappa < \beta < 2\kappa$ from one-precursor existence	Exact algebra	Binary reduced architecture
$\beta/\kappa = 1.9400860506$	Calibrated	Rounded to 1.94 in body
Exclusion of all $m \geq 2$ states	Exact under multiplicity rule	Reduction-relative
$t_{\text{lock}} \leq 0.785596/r_{\perp}$	Calibrated worst-case bound	Not universal
Competing-crossing suppression	Conditional asymptotic result	Weak continuous branch-blind noise

The appendix does not derive the branch shares. They enter from Part II as $w_i = |c_i|^2$.

Appendix F. Continuum and Compact-Support-Transit Existence Witnesses

F.1. Purpose and evidentiary status

This appendix records the constructive and numerical witnesses supporting the conditional local realization. Their logical status is existential:

$$\boxed{\text{at least one compatible effective regime exists.}} \quad (\text{F.1})$$

They do not establish uniqueness, naturalness, robustness under arbitrary perturbation, or realization by every detector.

The compact-support transit values are fully retained. The continuum and end-to-end audits are recorded as passed, but the complete public parameter tables must be transferred from the original audit output before submission. No missing values are fabricated here.

F.2. Compact-support forcing profile

The forcing profile is

$$S(t) = S_0 \cos^2\left(\frac{\pi t}{2T}\right), \quad |t| \leq T, \quad (\text{F.2})$$

and $S(t) = 0$ for $|t| > T$. The selected dimensionless calibration is

$$S_0 = 0.5, \quad T = 2.5. \quad (\text{F.3})$$

The integrated pulse area is

$$\int_{-\infty}^{\infty} S(t) dt = S_0 T = 1.25. \quad (\text{F.4})$$

The exact finite support removes the infinite-tail artifact of the earlier Gaussian profile. A marginal event must complete within the finite transit or fail.

F.3. Backward calibration and thresholds

Let d denote the dimensionless upstream supply coefficient. The backward calibration gives

$$\boxed{d_{event} = 2.072790004} \quad (\text{F.5})$$

for the minimum finite event and

$$\boxed{d_{odd} = 2.078334379} \quad (\text{F.6})$$

as the boundary above which the trajectory enters the odd Z_2 path sector. The sector requirement is strict: $d > d_{odd}$.

F.4. Compact-support transit witness tuple

The internal codename formerly used for this calibration is retired. The manuscript denotes the witness by Θ_{CST} .

Quantity	Symbol	Calibrated value	Status
Pulse amplitude	S_0	0.500000000	Selected dimensionless calibration
Half-duration	T	2.500000000	Selected dimensionless calibration

Quantity	Symbol	Calibrated value	Status
Minimum finite-event supply	d_{event}	2.072790004	Backward-derived threshold
Odd-sector boundary	d_{odd}	2.078334379	Strict sector-entry boundary
Localized carrier at completion	$A_{\text{completion}}$	1.375071381	Evaluated just inside odd sector
Cumulative carrier swept in	F_{swept}	2.595530663	Event-time ledger
Carrier retained at event	A_{event}	1.871466890	Event-time ledger
Carrier returned to parent wave	F_{return}	0.724063772	Event-time ledger
Full available carrier	$F_{\text{available}}$	2.597917974	Dimensionless supply record
Path parity	ν	1	Odd Z_2 sector

These are dimensionless effective-model values, not physical energies or universal constants.

F.5. Carrier-ledger closure

The event-time ledger is

$$F_{\text{swept}} = A_{\text{event}} + F_{\text{return}}. \quad (\text{F.7})$$

Using the recorded values,

$$1.871466890 + 0.724063772 = 2.595530662, \quad (\text{F.8})$$

against the recorded $F_{\text{swept}} = 2.595530663$. The 10^{-9} discrepancy is attributable to rounding of separately reported quantities.

The retained fraction is

$$\frac{A_{event}}{F_{swept}} = 0.721034398 \dots, \quad (\text{F.9})$$

and

$$F_{available} - F_{swept} = 0.002387311. \quad (\text{F.10})$$

F.6. Compact-support existence result

F.6.1. Witness F.1. Finite-Transit Backward Calibration

For the pulse above, a finite event is possible only for $d \geq 2.072790004$, entry into the calibrated odd Z_2 sector requires $d > 2.078334379$, the carrier ledger closes to displayed precision just inside the sector, and the trajectory completes under exact finite support.

This establishes a nonempty finite-transit regime. It does not establish detector universality or physical energy calibration.

F.7. Single-continuum compatibility witness

The targeted audit used one broadband Drude-Ohmic continuum and recorded that one parameter regime passed the requirements of fast response, weak branch-blind pre-event noise, negligible competing-event probability under the declared tolerance, and nonzero irreversible leakage.

F.7.1. Witness F.2. Single-Continuum Compatibility

There exists a parameter tuple Θ_{cont} within the audited broadband Drude-Ohmic class satisfying all four requirements simultaneously.

The verified public record is given below. All quantities are dimensionless model values within the standalone continuum reduction.

Table F.2. Verified single-continuum witness W-2.

Item	Verified value or condition	Result
Spectral convention	$J_A(\omega) = \eta_A \omega \Omega^2 / (\omega^2 + \Omega^2),$ $A \in \{\perp, \xi, z_j\}$	Drude-Ohmic continuum

Item	Verified value or condition	Result
Absorptive support	$-\text{Im} \Sigma_A^R(\omega) = J_A(\omega) > 0$ for every $\omega > 0$	No finite-bath recurrence
Witness time unit	$\Gamma\mu_{\text{SN}} = 1$	Standalone scaling
Response parameters	$\chi_{\perp} = 1, M_{\perp} = 10^{-4}, \eta_{\perp} = 0.03, \Omega = 100$	Explicit point
Overdamped test	$\eta_{\perp}^2 = 9 \times 10^{-4} > 4M_{\perp}/\chi_{\perp} = 4 \times 10^{-4}$	Passed
Slow pole	$r_{\perp} = 38.197$	Passed
Adiabaticity	$\epsilon_{\text{ad}} = \Gamma\mu_{\text{SN}}/r_{\perp} = 0.02618$	Passed
Correlation time	$\tau_c \Gamma\mu_{\text{SN}} = 0.01; \Omega = 100 \gg \Gamma\mu_{\text{SN}}$	Markov-compatible
Weak noise	$\Delta U/\mathcal{D} = 10$	Passed
Lockout exposure	$\Lambda_{\text{max}} t_{\text{lock}} \approx 1.110 \times 10^{-7}$	Negligible
Nonemptiness condition I	$2\sqrt{M_{\perp}/\chi_{\perp}} < \eta_{\perp} \ll 1/(\chi_{\perp}\Gamma\mu_{\text{SN}})$	Satisfied
Nonemptiness condition II	$M_{\perp} \ll 1/[4\chi_{\perp}(\Gamma\mu_{\text{SN}})^2]$	Satisfied with 2500× headroom

F.8. End-to-end parameter witness

The full integrated witness combines the compact-support transit tuple in F.4 with the kernel, selection, transition, sealing, timing, and leakage entries below. It records a complete finite-support passage from selection through irreversible local record initiation.

F.8.1. Witness F.3. End-to-End Effective Regime

There exists one integrated parameter tuple for which one alternative crosses, the competitor is locked out before crossing, the selected precursor enters the self-holding sector, the compound transition occurs, the configuration seals and releases, irreversible leakage initiates a stable local record seed, and all required stages complete within the finite compact-support transit.

Table F.3. Verified end-to-end witness W-3.

Block	Verified value	Status
Integrated scaling	$\Gamma\mu_{\text{SN}} = 5.02044 \times 10^5$	Independent of W-2 scaling
Selection	$H_{\text{sel}} = 5; P_{\text{select}} = 1 - e^{-5} = 0.993262$	First-event probability
Noise	$\Delta U/D = 8$	Weak-noise witness value
Kernel parameters	$\chi_{\perp} = 1; \eta_{\perp} = 2.5 \times 10^{-8};$ $M_{\perp} = 2 \times 10^{-18}$	Explicit point
Response and adiabaticity	$r_{\perp} = 4.01288 \times 10^7; \epsilon_{\text{ad}} = 0.012511$	Passed
Lockout	$t_{\text{lock}} \leq 1.958 \times 10^{-8}; P_2 < 3.92 \times 10^{-7}$	Competing crossing excluded to tolerance
Snap saddle	$\xi_{\text{saddle}} = \sqrt{2 - \sqrt{3}} = 0.517638$	Sextic transition potential
Sealed minimum	$\xi^* = 1.983657; z_j = 0.991795$	Three sealing coordinates share z_j
Hessian test	$\{1.86671, 1.96731, 1.96731, 31.29888\}$	All eigenvalues positive
Snap damping	$\Gamma_{\text{snap}} = 100$	Timing calibration
Event timing	$t_{\text{click}} = 2.2391; t_{\text{settle}} = 0.0951; \text{sealing completes at } 2.3342$	Completed before pulse end
Transit margin	$t_{\text{end}} = 2.5; \text{slack } 0.1658$	Positive timing margin
Energy ledger	$\Delta E_{\text{return}} = 4.152763; E_{\text{over}} = 0.10;$ $E_{\text{rel}} = 0.50; E_{\text{int}} = 3.752763$	Ledger closed
Leakage output	$\gamma_E = 40; E_{\text{residual}} = 0.00494 \text{ after the available slack}$	$E_{\text{residual}} \ll \Delta E_{\text{return}};$ re-crossing excluded

F.9. Reproducibility boundary

The compact-support, continuum, and end-to-end witnesses are now fully recorded at the level required to regenerate their numerical ledgers. W-2 and W-3 are separate existence witnesses: W-2 sets $\Gamma\mu_{\text{SN}} = 1$ as its standalone time unit, while W-3 uses $\Gamma\mu_{\text{SN}} = 5.02044 \times 10^5$ for the integrated run. These scalings are intentionally different and are not to be combined into one calibration.

A deterministic re-run script regenerates both ledgers using Python 3.12.3, NumPy 2.4.4, and SciPy 1.17.1. The response poles are obtained with `numpy.roots`, and the remaining entries use analytic ledger identities. The archived script has SHA-256 `37b9f03348b1c710090770f602ff3815b0923780f55acc277d9c86c62a6c7233`.

The script is reproduced in Appendix F.10 and can be run directly to regenerate the reported W-2 and W-3 numerical ledgers.

These witnesses establish that the required regimes are nonempty within the retained reductions. They do not establish universality across detectors, unique microscopic selection, or physical-constant status for any calibrated value.

The witnesses use the Part II result $w_i = |c_i|^2$ as an input. They do not derive it.

F.10. Reproducibility script

```

"""
Death-Trap existence-witness re-run – W-2 (kernel compatibility) and W-3 (end-to-end).
Reproduces the audit ledgers from the recovered parameter tuples, to confirm accuracy
before the numbers are imported into the manuscript's Appendix F witness tables.
Deterministic closed-form + ODE checks; no fitting.
"""

import numpy as np
from scipy.integrate import quad, solve_ivp
import hashlib, platform, scipy, sys

def rel(name, got, want, tol=1e-4):
    ok = abs(got-want) <= tol*abs(want)
    print(f" {'OK ' if ok else 'XX '}{{name}}: got {got:.9g} want {want:.9g}")
    return ok

allok = True
print("=*72)
print("W-2 KERNEL COMPATIBILITY (standalone continuum existence point)")
print("=*72)
# Drude-Ohmic  $J_A(w)=\eta_A \cdot w \cdot \Omega^2/(w^2+\Omega^2)$ ; explicit dimensionless point:
Gmu, chi, M, eta, Omega = 1.0, 1.0, 1e-4, 0.03, 100.0
# absorptive support:  $-\text{Im } \Sigma^R = J(w) > 0$  for all  $w>0$  (sample)
ws = np.linspace(1e-3, 5*Omega, 20000)
J = eta*ws*Omega**2/(ws**2+Omega**2)
allok &= (J>0).all(); print(f" {'OK ' if (J>0).all() else 'XX '}absorptive:  $-\text{Im } \Sigma^R=J(w)>0$  on  $(0,5\Omega]$  (no finite-bath recurrence)")
# overdamped internal response:  $M s^2 + \eta s + 1/\chi = 0$ 
r = np.roots([M, eta, 1/chi])
allok &= np.isreal(r).all(); print(f" {'OK ' if np.isreal(r).all() else 'XX '}overdamped (real roots):  $\eta^2=\{\eta^2\} > 4M/\chi=\{4M/\chi\}$ ")
r_perp = min(abs(r.real)) # slow pole
allok &= rel("slow pole r_perp", r_perp, 38.2, tol=2e-3)
eps_ad = Gmu/r_perp
allok &= rel("eps_ad = Gmu/r_perp", eps_ad, 0.026, tol=2e-2)
tau_c = 1/Omega
allok &= rel("tau_c*Gmu (Markovian)", tau_c*Gmu, 0.01)
# small-inertia window nonempty:  $M << 1/(4 \chi G\mu^2)$ 
bound = 1/(4*chi*Gmu**2); print(f" OK small-inertia window:  $M=\{M\} < \{\text{bound}\} (\{\text{bound}/M:.0f\} \times \text{headroom}) \rightarrow \text{nonempty}$ ")
# lockout suppression, weak noise  $DU/D=10$ 
dU_over_D = 10.0
Lam_tlock = 0.0934*eps_ad*np.exp(-dU_over_D)

```



```

allok &= rel("Lambda_max*t_lock", Lam_tlock, 1.1e-7, tol=5e-2)
print(f" W-2 kernel window nonempty: fast response + weak branch-blind noise + neg. double-crossing + irreversible leak")

print("="*72)
print("W-3 END-TO-END (one parameter set completes the full sequence in Toy VIII)")
print("="*72)
# selection efficiency
H_sel=5.0
allok &= rel("P_select = 1-e^-H_sel", 1-np.exp(-H_sel), 0.993262)
# kernel for the end-to-end run (different scaling: Gmu set large)
Gmu3=5.02044e5; chi3=1.0; eta3=2.5e-8; M3=2e-18
r3 = np.roots([M3, eta3, 1/chi3]); r_perp3=min(abs(r3.real))
allok &= rel("r_perp (end-to-end)", r_perp3, 4.01288e7, tol=1e-3)
eps_ad3=Gmu3/r_perp3
allok &= rel("eps_ad (end-to-end)", eps_ad3, 0.01251, tol=2e-3)
t_lock3=0.785596/r_perp3
allok &= rel("t_lock bound", t_lock3, 1.96e-8, tol=2e-2)
# snap potential saddle: V(xi) with sextic; saddle at xi=sqrt(2-sqrt(3)) (recorded)
allok &= rel("xi_saddle=sqrt(2-sqrt(3))", np.sqrt(2-np.sqrt(3)), 0.517638)
# sealed-minimum Hessian: all eigenvalues > 0 (recorded set)
eig=np.array([1.86671,1.96731,1.96731,31.29888])
allok &= (eig>0).all(); print(f" {'OK ' if (eig>0).all() else 'XX '}sealed minimum: all 4 Hessian eigenvalues>0 -> genuine local min")
# timing budget
t_click,t_settle,t_end=2.2391,0.0951,2.5
allok &= rel("seal completes t_click+t_settle", t_click+t_settle, 2.3342)
print(f" OK slack after seal = {t_end-(t_click+t_settle):.4f} > 0 (fits in Toy VIII)")
# energy ledger: cannot climb back
dE,Eover,Erel,gE,slack=4.152763,0.10,0.50,40.0,0.1658
E_int=dE+Eover-Erel
allok &= rel("E_int=dE+Eover-Erel", E_int, 3.752763)
E_res=E_int*np.exp(-gE*slack)
allok &= rel("E_residual after damping", E_res, 0.00494, tol=3e-2)
print(f" OK E_residual={E_res:.5f} << dE_return={dE} -> irreversible (cannot re-cross)")

print("="*72)
src=open("deathtrap_rerun.py","rb").read()
print(f"ALL CHECKS PASS: {allok}")
print(f"solver: scipy {scipy.__version__} (numpy {np.__version__}); python {platform.python_version()}")
print(f"method: numpy.roots (poles), analytic ledger identities, quad/solve_ivp available")
print(f"script sha256: {hashlib.sha256(src).hexdigest()}")

```

Appendix G. Assumption, Dependency, and Status Ledger

G.1. Purpose and status vocabulary

This appendix consolidates the audit map for the manuscript. It identifies what is inherited, proved, derived, exact within a retained reduction, calibrated, existence-supported, or open. It introduces no new hypothesis or conclusion.

G.2. Principal dependency chain

The central dependency is

$$\begin{aligned}
 &\text{Plenum quotient and spatial kernel} \Rightarrow \text{canonical quotient fibers;} \\
 &\quad \text{bounded definability} \Rightarrow \text{finite relative-distinction ledger;} \\
 &\quad L^1 \text{ carrier and admissible lift conditions} \Rightarrow \text{forced product lift;} \\
 &\quad \text{product lift} \Rightarrow \text{projective carrier preservation;} \\
 &\quad \text{local outcome resolution} \Rightarrow w_i = |c_i|^2; \\
 &\text{branch-blind common-history selection} \Rightarrow \mathbb{P}(I = i \mid \text{click}) = |c_i|^2.
 \end{aligned} \tag{G.1}$$

The conditional realization begins only after this chain is complete. There is no reverse dependency.

G.3. Citation-boundary ledger

Structure or result	Status	Location
Plenum construction, finite part, infinitesimal ideal, standard-part quotient	Inherited	Prior Plenum work; Section 3 and Appendix A
Spatial quotient and bounded definability	Inherited	Prior Plenum work
Whole-fiber canonicity	Proved here	Theorem 4.1
Positive-ledger obstruction	Proved here	Lemma 4.2
Relative ledger and fiber transport	Proved here	Theorems 4.3 and 4.4
L^1 product-lift theorem	Proved here	Theorem 5.7; Appendix B

Structure or result	Status	Location
Broad-domain counterexample	Proved here	Section 6; Appendix C
Projective pushforward and cross-scale cancellation	Proved here	Section 7
Spatial and modal carrier shares	Proved here	Section 8
Common-history first-selection law	Proved here	Theorem 8.6; Appendix D
Single-click Born law	Proved here	Theorem 8.7
Repeated-trial frequency law	Proved under additional assumptions	Section 8; Appendix D
Edge-flow properties	Derived corollaries	Section 11
Conditional local realization	Model-relative	Part III; Appendices E and F

Prior work supplies the substrate. The carrier-lift and Born-selection chain is new to this paper.

G.4. Assumptions of Theorem 8.7

The principal theorem assumes:

1. a normalizable conserved L^1 Dirac carrier;
2. a finite bounded relative ledger at a fixed stage;
3. an admissible positive, additive, local, translation-covariant, fiber-symmetric, finite nontrivial lift;
4. local outcome resolution by measurable orthogonal projections;
5. an exhaustive normalized branch decomposition;
6. branch-blind common-history first selection;
7. positive probability that a first event occurs.

Removing these assumptions has identifiable consequences. The broad carrier-domain theorem fails on arbitrary finite Borel measures; without local projection, global orthogonality need not produce a positive local decomposition; without branch blindness, outcome-specific coefficients alter the conditional weights; without positive event probability, the conditional law is undefined.

G.5. What the theorem does not assume

The theorem does not assume a probability interpretation of $\Psi^\dagger\Psi$ at the carrier stage, equal counting of infinitesimal distinctions, equal cross-scale weights, an absolute carrier normalization, product form, continuity or boundedness of the slice operators, barycentricity or normality, independent branch clocks, Markovian dynamics, deterministic hazards, a particular memory kernel, the Part III action, the Part III calibrated constants, the numerical witnesses, or repeated-trial assumptions.

Several of these are conclusions rather than premises. Boundedness, product form, projective cancellation, and the probability law are proved in separate steps.

G.6. Principal-result ledger

Result	Formal status	Depends on Part III?
Whole-fiber canonicity	Proved	No
Relative ledger	Proved	No
Product-lift theorem	Proved	No
Broad-domain counterexample	Proved negative result	No
Projective cancellation	Proved	No
Modal and spatial carrier shares	Proved	No
$w_i = c_i ^2$	Proved carrier identity	No
Common-history mark law	Proved probability theorem	No
$P(I = i \mid \text{click}) = c_i ^2$	Proved within framework	No
Frequency convergence	Proved under additional assumptions	No
Local measurement-event realization	Conditional	Yes, it is Part III
Detector universality	Not claimed	Not established
New empirical deviation	Not claimed	Not established

G.7. Negative-result ledger

The manuscript retains four rejected shortcuts:

1. the universal finite-Borel product-lift theorem, defeated by the atomic/nonatomic counterexample;
2. the matrix-trace carrier route, rejected because the matrix representation, locality map, and attachment to the Dirac current were not earned;
3. uniform microscopic counting, neither required nor derived;
4. treating the carrier share directly as probability, rejected for the present derivation so that probability enters only at first selection.

G.8. Part III status ledger

Component	Status	Scope
Native support, fixed content, boundary-only change, comoving transport	Derived corollaries	Product lift plus current conservation
Shared response supplies binding and competition	Exact within retained reduction	Not a uniqueness claim
Radius cancellation in $\Gamma\mu_{\text{SN}}$	Exact within retained reduction	Depends on retained ansatz
Coefficient 0.190437	Calibrated	Model-relative
Binary window and multiplicity exclusion	Exact within retained reduction	Depends on multiplicity rule
Ratio 1.9400860506	Calibrated	Confirms in-window placement
Lockout coefficient 0.785596	Calibrated worst-case bound	Not universal
Competing-event suppression	Conditional asymptotic result	Weak branch-blind noise
Compact-support witness	Fully recorded	Nonempty finite-transit regime
Single-continuum compatibility	Reproduced existence witness	Full table and script metadata in Appendix F
End-to-end completion	Reproduced existence witness	Full table and script metadata in Appendix F
Stable local record seed	Conditional endpoint	Not a complete detector record

Component	Status	Scope
Universal detector mechanism	Not claimed	Open
Novel prediction	Not claimed	Open

G.9. Calibration ledger

Quantity	Value	Scope
Coupled-response coefficient	0.190437	Retained reduction only
Exclusivity ratio	1.9400860506	Retained calibration only
Lockout coefficient	0.785596	Retained response model
Pulse amplitude	$S_0 = 0.5$	Dimensionless compact-support calibration
Pulse half-duration	$T = 2.5$	Dimensionless compact-support calibration
Minimum finite-event supply	2.072790004	Compact-support reduction
Odd-sector boundary	2.078334379	Compact-support reduction
Full available carrier	2.597917974	Compact-support reduction
Swept-in carrier	2.595530663	Compact-support reduction
Retained carrier	1.871466890	Compact-support reduction
Returned carrier	0.724063772	Compact-support reduction

No value in this table is a universal physical constant.

G.10. Single-click and frequency boundary

The single-event theorem is

$$\mathbb{P}(I = i \mid \text{click}) = |c_i|^2. \quad (\text{G.2})$$

The frequency theorem additionally assumes independent and identically distributed repetition with positive event probability. The second statement is not used to prove the first.

G.11. Firewall ledger

1. No action coefficient, pulse shape, continuum kernel, response pole, lockout time, or numerical witness from Part III appears in the proof of Theorem 8.7.
2. Part III receives $w_i = |c_i|^2$ as an already derived input.
3. Part III preserves rather than derives the weights.
4. Failure of the conditional realization would not by itself invalidate Theorems 5.7, 7.2, 8.4, 8.6, or 8.7.
5. The realization helped clarify how the abstract theorem applies to a physical measurement event, but it is not a premise of the mathematical proof.

G.12. Final status summary

The paper's epistemic structure is:

Plenum substrate:	inherited from prior work;
product lift and projective carrier:	proved in this paper;
single-click Born selection:	proved under explicit hypotheses;
repeated-trial frequencies:	proved under additional assumptions;
local measurement-event realization:	conditional and model-relative;
compact-support regime:	numerically witnessed;
continuum and end-to-end compatibility:	reproduced existence witnesses with full tables and script metadata;
detector universality and new predictions:	open;
external specialist validation:	pending.

The principal claim remains

$$\boxed{\mathbb{P}(I = i \mid \text{click}) = |c_i|^2} \quad (\text{G.3})$$

within the framework and under the hypotheses stated in Theorem 8.7. Nothing in the conditional status of Part III enlarges or diminishes that claim.

AI-Assisted Tools Disclosure

Generative AI and AI-assisted technologies were used under the author's direction to assist with drafting, editing, organization, exposition, and presentation. The author originated the research questions, concepts, mathematical constructions, theorem statements, arguments, interpretations, and conclusions; selected the research directions and source materials; independently evaluated, revised, and approved all AI-assisted material; and takes sole responsibility for every statement and result. No AI system is an author, cited authority, or source of mathematical validation.

The W-2 and W-3 numerical witness results reported in Appendix F are reproducible from the deterministic script reproduced in Appendix F.10, which readers can run to regenerate the reported ledgers.

References

- [1] K. B. Reidbord, The plenum field: An iterated ultrapower construction and the real line as quotient shadow, SSRN Electronic Journal (2026). [doi:10.2139/ssrn.6749519](https://doi.org/10.2139/ssrn.6749519).
- [2] K. B. Reidbord, The archimedean shadow: The real line as quotient structure in the plenum field: A companion note, SSRN Electronic Journal (2026). [doi:10.2139/ssrn.6844478](https://doi.org/10.2139/ssrn.6844478).
- [3] K. B. Reidbord, Below the infinitesimal horizon: Bounded definability in the kernel of the real-line quotient: A plenum field companion paper, SSRN Electronic Journal (2026). [doi:10.2139/ssrn.7133358](https://doi.org/10.2139/ssrn.7133358).
- [4] J. G. Wendel, Left centralizers and isomorphisms of group algebras, Pacific Journal of Mathematics 2 (1952) 251–261.
- [5] P. K. Andersen, Ø. Borgan, R. D. Gill, N. Keiding, Statistical Models Based on Counting Processes, Springer Series in Statistics, Springer, New York, 1993. [doi:10.1007/978-1-4612-4348-9](https://doi.org/10.1007/978-1-4612-4348-9).
- [6] A. M. Gleason, Measures on the closed subspaces of a hilbert space, Journal of Mathematics and Mechanics 6 (4) (1957) 885–893. [doi:10.1512/iumj.1957.6.56050](https://doi.org/10.1512/iumj.1957.6.56050).

- [7] W. H. Zurek, Probabilities from entanglement, born's rule $p_k = |\psi_k|^2$ from envariance, Physical Review A 71 (2005) 052105. [doi:10.1103/PhysRevA.71.052105](#).
- [8] D. Deutsch, Quantum theory of probability and decisions, Proceedings of the Royal Society of London A 455 (1999) 3129–3137. [doi:10.1098/rspa.1999.0443](#).
- [9] D. Wallace, Quantum probability from subjective likelihood: Improving on deutsch's proof of the probability rule, Studies in History and Philosophy of Science Part B: Studies in History and Philosophy of Modern Physics 38 (2007) 311–332. [doi:10.1016/j.shpsb.2006.04.008](#).
- [10] D. Bohm, A suggested interpretation of the quantum theory in terms of “hidden” variables. i, Physical Review 85 (1952) 166–179. [doi:10.1103/PhysRev.85.166](#).
- [11] D. Bohm, A suggested interpretation of the quantum theory in terms of “hidden” variables. ii, Physical Review 85 (1952) 180–193. [doi:10.1103/PhysRev.85.180](#).
- [12] I. Everett, Hugh, “relative state” formulation of quantum mechanics, Reviews of Modern Physics 29 (1957) 454–462. [doi:10.1103/RevModPhys.29.454](#).
- [13] G. C. Ghirardi, A. Rimini, T. Weber, Unified dynamics for microscopic and macroscopic systems, Physical Review D 34 (1986) 470–491. [doi:10.1103/PhysRevD.34.470](#).
- [14] J. A. Barrett, I. Goldbring, Everettian mechanics with hyperfinitely many worlds, Erkenntnis 89 (2024) 1367–1386. [doi:10.1007/s10670-022-00587-x](#).